

Spectral Sequences of Flow Categories

Arye Deutsch*

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Abstract

Framed flow categories and their stable homotopy types are central tools in modern Floer, contact, Khovanov, and knot homotopy theories. In this paper, given an $\mathcal{X}$ -structured flow category $\mathcal{C}$ and a generalized homology theory A equipped with a map $M\mathcal{X} \rightarrow A$, we construct an Atiyah-Hirzebruch-type spectral sequence which converges to the homology of the realization $A_*(|\mathcal{C}|)$. Moreover, if A is a bordism theory - for example, if $\mathcal{C}$ is framed and A is stable homotopy - we provide a description of the higher pages and differentials in terms of gluing the flow manifolds in $\mathcal{C}$ to a geometric analog of Toda brackets, an idea traced back to Floer[13].

This work provides a bordism-level perspective on spectral sequences, aligning with the approach of Abouzaid and Blumberg, who developed a bordism model for spectra. We demonstrate these spectral sequence techniques by calculating the unoriented and oriented bordism homology of real projective spaces using Morse flow data.

Independent of flow categories, we provide a description of the spectral sequence of a chain complex in a stable category in terms of morphisms between chain complexes.

*Einstein Institute of Mathematics, Hebrew University of Jerusalem, Israel. arye.deutsch@mail.huji.ac.il

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1 Introduction

[27]

Short Introduction

Flow categories originally emerged as a categorical bookkeeping mechanism to organize the geometric and analytic data extracted from Hamiltonian systems on a manifold [10]. In recent years, flow categories and their stable homotopy types have become central tools for lifting invariants in Floer, contact, and Khovanov homology into the realm of stable homotopy theory. A natural question arises: how can we systematically compute the generalized homology of these realization spectra directly from the underlying geometric flow data?

Here is an intuitive description of the main theorem of the paper using a toy model coming from Morse theory. Let M be a closed manifold and $f : M \rightarrow \mathbb{R}$ a Smale Morse function. Denote by $\text{Crit}(f)$ the set of critical points of f , and by $\text{ind} : \text{Crit}(f) \rightarrow \mathbb{Z}$ the index function. For simplicity, assume that there is one critical point for each index $i \in \{0, \dots, n\}$, and denote the critical points by x_i where $\text{ind}(x_i) = i$. The first result is:

There is a spectral sequence

$$E_1^{n,p} = \Omega_{n-p}^{\text{fr}} \implies \pi_n \Sigma_+^\infty M.$$

where Ω_*^{fr} is the framed bordism ring, and $\Sigma_+^\infty M$ is the stable homotopy type of M .

Let x_j, x_i be two critical points of f with $j > i$. We denote by $\mathcal{M}(x_j, x_i)$ the compactification of the moduli space of flow lines from x_j to x_i , up to reparametrization. Recall that $\mathcal{M}(x_j, x_i)$ is a framed manifold with corners of dimension $\text{ind}(x_j) - \text{ind}(x_i) - 1$, and that the boundary structure is given by

$$\partial \mathcal{M}(x_j, x_i) = \bigcup_{j > r > i} \mathcal{M}(x_j, x_r) \times \mathcal{M}(x_r, x_i).$$

Let W_p be a representative of a bordism class in $E_1^{n,p} = \Omega_{n-p}^{\text{fr}}$. We have that W_p survives to the $(r+1)$ -th page (also denoted by $W_p \in Z_{r+1}$) if and only if there is a family of framed manifolds $\{W_s\}$ for $s \in \{p-1, \dots, p-r\}$ of dimension $n-s$ such that the corner structure is given by

$$\partial W_s = \bigcup_{p \geq t > s} W_t \times \mathcal{M}(x_t, x_s).$$

We can then glue the collection $W_s \times \mathcal{M}(x_s, x_{p-(r+1)})$ together. Let $t, u \in \mathbb{Z}$ be such that $p \geq t > u \geq p - (r+1)$. By the definition of the boundary structure of W_u and $\mathcal{M}(x_t, x_{p-(r+1)})$, we have that

$$W_t \times \mathcal{M}(x_t, x_{p-(r+1)}) \leftarrow W_t \times \mathcal{M}(x_t, x_u) \times \mathcal{M}(x_u, x_{p-(r+1)}) \hookrightarrow W_u \times \mathcal{M}(x_u, x_{p-(r+1)})$$

and these boundary components are shared only by these two manifolds of the collection $W_s \times \mathcal{M}(x_s, x_{p-(r+1)})$. On the other hand, all the boundary components of $W_s \times \mathcal{M}(x_s, x_{p-(r+1)})$ are of this form, so we can glue the collection $W_s \times \mathcal{M}(x_s, x_{p-(r+1)})$ together to obtain a framed closed manifold which we denote by

$$\mathcal{M}(-, x_{p-(r+1)}) \circ W := \bigcup_{s \in \{p, \dots, p-r\}} \mathcal{M}(x_s, x_{p-(r+1)}) \times W_s,$$

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which is an analog of the Toda bracket construction. Note that

$$\dim(\mathcal{M}(x_s, x_{p-(r+1)}) \circ W) = \dim(W_s) + \dim(\mathcal{M}(x_s, x_{p-(r+1)})) = n - s + (s - (p - (r + 1)) - 1) = n - p + r.$$

The last part of the theorem states that if $W_p \in E_1^{n,p} = \Omega_{n-p}^{\text{fr}}$ survives to Z_{r+1} , then the $r+1$ differential on Z_{r+1} is given by the bordism class of the Toda manifold:

$$d_{r+1}(W_p) = [\mathcal{M}(-, x_{p-(r+1)}) \circ W] \in \Omega_{n-p+r}^{\text{fr}} = E_1^{n-1, p-(r+1)}.$$

See Fig. 2 for an example of the structure of the family W_s and the flow manifolds, Fig. 3 for the gluing diagram of the Toda manifold, and Fig. 4 for the resulting closed manifold.

Background

Morse theory basics. The field of differential topology studies the connection between analysis on a smooth n -manifold M and its underlying topology. This field was pioneered by Morse in the beginning of the 20th century, and further developed by Smale, Thom, Milnor and more (see [28] for Morse first work and [26] for Milnor’s classic exposition of the subject). To begin understanding M , we take a “nice” (namely a Smale-Morse) function $f: M \rightarrow \mathbb{R}$. We extract the *critical points* of f and, for any critical point p , assign its *index* $\text{ind}(p)$ - which is defined as the signature of the Hessian of $-\nabla f$. Pictorially, this index represents the number of ‘downward’ axes at p , see Fig. 1.

For any two critical points y and x , we define $M(y, x)$ to be the moduli space of curves (up to reparametrization) in M that start at y and end at x , and follow the gradient field $-\nabla f$. We call such curves *flow lines*; see Fig. 1. Under generic conditions, $M(y, x)$ is a smooth (not necessarily closed) manifold of dimension $\text{ind}(y) - \text{ind}(x) - 1$.

Framed flow manifolds. Our main interests are *flow manifolds*, which are the compactification

$$\mathcal{M}(y, x) := \overline{M(y, x)}.$$

The compactification is given by adding *broken flow lines*, which are piecewise smooth curves that start at y and end at x and pass through other critical points. We get that the flow manifolds have a *corner structure*

$$\partial \mathcal{M}(z, x) = \bigcup_{\text{ind}(z) > \text{ind}(y) > \text{ind}(x)} \mathcal{M}(z, y) \times \mathcal{M}(y, x).$$

see Fig. 1. The fact that $\mathcal{M}(y, x)$ is constructed from the transverse intersection of two discs (one for curves ascending to y and one for curves descending to x) gives a stable *framing* on $\mathcal{M}(y, x)$ - a trivialization of the form

$$T\mathcal{M}(y, x) \oplus \mathbb{R}^{\ell_1} \simeq \mathbb{R}^{\ell_2}$$

where $\mathbb{R}^{\ell_i}$ are trivial vector bundles over $\mathcal{M}(y, x)$. Moreover, the framing of all the flow manifolds is compatible with the corners structure.

Morse homology. It’s worth saying that this data is sufficient to calculate the homology $H_*(M, \mathbb{Z})$. It is given as the homology of the Morse chain complex (C_*, d) . The generators of C_p are critical points with index p . Given two critical points y, x with $\text{ind}(y) - 1 = \text{ind}(x)$, the flow manifold $\mathcal{M}(y, x)$ is in fact compact; the coefficient of x in $d(y)$ is given by the signed count

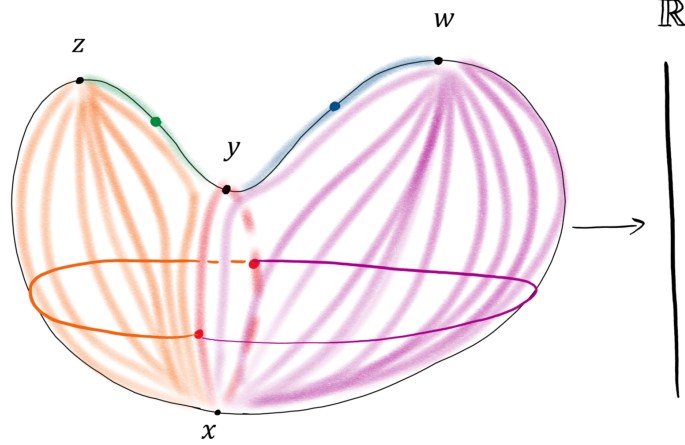


Figure 1: A Smale-Morse function on the sphere, given as projection to the z axis of the heart-shaped embedding. The letters represent critical points (with indices 2,2,1,0), the flow lines are drawn as soft, colorful lines, the solid colorful dots are representatives for the zero-dimensional flow manifolds and the solid colorful lines are representatives for the one-dimensional flow manifolds (notice the boundary structure as broken flow lines)

of $\mathcal{M}(y, x)$, where the signs are determined by the framing. The boundary structure of the one-dimensional moduli spaces $\mathcal{M}(z, x)$ shows that $d^2 = 0$. See [34].

Bordism and generalized homology. The fact that the flow manifolds are framed hints at an important theorem in differential topology, the Pontryagin-Thom isomorphism. We say that two closed framed n -manifolds M_1 and M_2 are *bordant* if there is a framed $(n + 1)$ -manifold W , with $\partial W = M_1 \sqcup M_2$, in a way that is compatible with the framing. We denote by Ω_n^{fr} the set of n dimensional framed manifolds up to bordism. In fact the collection Ω_*^{fr} form a graded ring, by disjoint union and Cartesian product.

The *Pontryagin-Thom isomorphism* states that there is an isomorphism

$$\Omega_n^{\text{fr}} \simeq \text{colim}_k [S^{n+k}, S^k]$$

between bordism groups of framed manifolds and stable homotopy groups of spheres (see [29, 35]). The right hand side is also known as the coefficient ring $\pi_n(\mathbb{S})$, where $\mathbb{S}$ is the spectrum representing stable homotopy. We can view the Morse chain complex as a chain complex with coefficient ring $\mathbb{Z} = \Omega_0^{\text{fr}} = \mathbb{S}_{\leq 0}$ and the differential is given by multiplication by the 0 dimensional flow manifolds $\mathcal{M}(y, x)$.

Floer theory A related question is the analysis of real-valued functions on objects that are not finite-dimensional closed manifolds. Floer developed this idea in the 1980s, leading to what is now known as Floer theory. In general, Floer’s idea is to generate a chain complex in which the differentials are defined by ‘counting solutions’ to a moduli problem (see [12] for initial formulations of this idea). In this case, the outcome does not yield the homology of a manifold, and one may ask: *Are Floer homology groups merely the shadow of a more structured object?*

Flow categories. The development of a framework to address these questions led Cohen, Jones, and Segal (CJS) in [10] to introduce *framed flow categories*, which axiomatize the data extracted from a Smale-Morse function on a manifold. In short, a framed flow category $\mathcal{C}$ consists of a set of objects (corresponding to critical points) with a grading function $|\cdot|$ (corresponding

to the index). For two objects $y, x \in \mathcal{C}$, the morphism space from y to x is an $(|y| - |x| - 1)$ -dimensional manifold with corners, denoted $\mathcal{M}_{y,x}$, equipped with a choice of trivialization of $T\mathcal{M}_{y,x} \oplus \mathbb{R}^k$ for some k (corresponding to the stable framing). Moreover, the composition is given by identifying the faces in the following way:

$$\mathcal{M}_{z,y} \times \mathcal{M}_{y,x} \hookrightarrow \mathcal{M}_{z,x}.$$

See [Section 3.3](#) for definitions and details. Another building block in the theory are *flow modules*. Given a flow category $\mathcal{C}$, a *right flow module* W over $\mathcal{C}$ (see [Definition 3.24](#) for the formal definition) consists of a collection of manifolds $\{W_x\}_{x \in \mathcal{C}}$, together with corner structure maps

$$W_y \times \mathcal{M}_{y,x} \hookrightarrow W_x.$$

A right flow module is of *degree* i if

$$\dim W_x = i - |x|$$

for each pair of objects y and x in $\mathcal{C}$. See [\[30\]](#) definition 4.6, 4.7. By abuse of notation, we denote the collection of right modules of degree i over $\mathcal{C}$

$$[*[i], \mathcal{C}].$$

Similarly, we can define left flow modules of degree i over $\mathcal{C}$ (see [Definition 3.25](#) for the formal definition) to be collections of manifolds $\{V_x\}_{x \in \mathcal{C}}$, together with corner structure maps

$$\mathcal{M}_{y,x} \times V_x \hookrightarrow V_y.$$

such that $\dim V_x = |x| - i$. We denote the collection of left modules of degree i over $\mathcal{C}$ by

$$[\mathcal{C}, *[i]].$$

A useful example for left module on a category $\mathcal{C}$ is given by fixing $x' \in \mathcal{C}$ and taking the collection

$$\{\mathcal{M}_{x,x'}\}_{x \in \mathcal{C}}. \tag{1}$$

Indeed we have that the corner structure is given by the maps

$$\mathcal{M}_{y,x} \times \mathcal{M}_{x,x'} \hookrightarrow \mathcal{M}_{y,x'}$$

The degree of this module is given by

$$|x| - \dim(V_x) = |x| - (|x| - |x'| - 1) = |x'| + 1$$

and we denote it simply by $\mathcal{M}_{-,x'}$.

Filtrations and coherent chain complexes. Another result of CJS was to attach a stable homotopy type to a framed flow category by the following idea.

Given a filtered space $X_0 \rightarrow X_1 \rightarrow \dots \rightarrow X_n = X$ we can take the successive cofibers to produce a “chain complex” defined by

$$K_* := X_n/X_{n-1} \xrightarrow{d} \Sigma(X_{n-1}/X_{n-2}) \xrightarrow{d} \dots \xrightarrow{d} \Sigma^n X_0,$$

where d^2 is null-homotopic via the canonical null homotopy arising from X . Moreover, viewing K_* from higher-categorical perspective allows us to encode additional data. In particular, there exist two null homotopies for d^3 that can be concatenated to form a loop in a suitable mapping space; the chain complex is then equipped with a null homotopy of this loop, and so forth. We call this structure a *coherent chain complex*. We can attempt to recover X by taking *iterated cofibers*. For example, note that

$$\Sigma^2 X_1 = \text{cof}(\Sigma^1 X_1/X_0 \rightarrow \Sigma^2 X_0), \quad \Sigma^2 X_2 = \text{cof}(\text{cof}(X_2/X_1 \rightarrow \Sigma X_1/X_0) \rightarrow \Sigma^2 X_0)$$

and indeed, we can recover X up to a certain suspension. See [Section 2.1](#) and [\[5\]](#) for full detailed text.

Building on this idea, CJS showed that a framed flow category $\mathcal{C}$ can be collapsed, via a Pontryagin-Thom-type construction, into a coherent chain complex K in Spectra. After taking iterated cofiber, we obtain a spectrum

$$|\mathcal{C}| := \lim_p \text{cof}(\text{cof}(\dots \text{cof}(\text{cof}(K_p \rightarrow K_{p-1}) \rightarrow K_{p-2}) \rightarrow \dots \rightarrow K_1) \rightarrow K_0)$$

which is called the *realization* of $\mathcal{C}$.

Remark. Note that $|\mathcal{C}|$ depends on the choice of framing for the manifolds $M_{b,a}$.

Example - Morse flow category. To combine the ideas of framed flow categories and chain complexes, let $f : M \rightarrow \mathbb{R}$ be a manifold equipped with a Smale-Morse function. First, we construct from f a framed flow category $\mathcal{C}_f$ - by taking $\mathcal{M}_{y,x} := \mathcal{M}(y,x)$, then we collapse it to a chain complex $K_{\mathcal{C}_f}$ and realize it as $|\mathcal{C}_f| \in \mathbf{Sp}$. A result by CJS [\[10\]](#), states that this procedure recovers the stable homotopy type of M :

$$|\mathcal{C}_f| \simeq \Sigma_+^\infty M.$$

This framework has proven useful in Floer theory, where the solutions to the moduli problem sometimes exhibit the structure of a flow category $\mathcal{C}$. In such cases, we can realize the flow category, demonstrating that Floer homology is indeed a shadow of $|\mathcal{C}|$.

Looking back to Morse homology, we observe that the homology of a flow category can be determined from its zero-dimensional morphism manifolds. The fact that we can recover the stable homotopy type of M from its framed flow category motivates the idea that we can read off the (generalized) homology of M using the flow manifolds.

This idea goes back to Floer, who pointed out in [\[13\]](#) that: If h_* is a general homology theory, trajectory space analysis suggests contributions from compact components of $\mathcal{M}(x,y)$. For stable homotopy $h_* = \pi_*^s$, this corresponds to the framed cobordism type. While higher-dimensional components of $\mathcal{M}(x,y)$ may not be compact, a modified approach could yield a spectral sequence. This framework has limited use in finite-dimensional Morse theory but may apply to infinite-dimensional settings.

Gluing it together. *In this paper, given a flow category, we use the framework of CJS, Lurie, and Ariotta-Krause to generalize Floer's idea and construct an Atiyah-Hirzebruch-like spectral sequence. This spectral sequence is given explicitly by gluing the flow manifolds.*

One can think of this result as the generalization of Morse and Floer homology in the sense that we provide a recipe for starting with a manifold or Floer data and reading off its generalized homology groups using the geometric and analytic data.

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Notations

We denote by **Top** the 1-category of topological spaces, by **Spaces** the category of homotopy types (or ∞ -groupoids), and **Sp** is the category of spectra. The stabilization functor is denoted $\Sigma_+^\infty : \mathbf{Spaces} \rightarrow \mathbf{Sp}$, and $\mathrm{Ch}(\mathbf{Sp})$ and $\mathrm{Fil}(\mathbf{Sp})$ denote the categories of coherent chain complexes in Spectra and (complete) filtered Spectra, respectively.

For a topological space Y , we denote by Y_+ its one-point compactification (if Y is compact, then $Y_+ = Y \cup \{*\}$). For a CW complex Y , we denote by $\mathrm{sk}_n Y$ its n -th skeleton.

For a stable tangential structure $\xi : \mathcal{X} \rightarrow BO$, we denote by $\Omega_*^\mathcal{X}$ the graded $\mathcal{X}$ -bordism ring and by $M\mathcal{X}$ the Thom spectrum associated with the $\mathcal{X}$ -structure (see [Section 7](#) for definitions).

In this paper, a manifold is assumed to be a *closed manifold* unless stated otherwise.

Let $\mathcal{C}$ be a flow category. For $q, p \in \mathbb{Z}$ we denote by $\mathcal{C}_{[q,p]}$ the full subcategory of $\mathcal{C}$ consisting of objects satisfying $q \geq |x| \geq p$, and we define $\mathcal{C}_p := \mathcal{C}_{[p,p]}$. For brevity, we write

$$M_{p,q} := \bigsqcup_{y \in \mathcal{C}_q, x \in \mathcal{C}_p} M_{y,x}.$$

for the union of the flow manifolds, and for a right / left module over $\mathcal{C}$ we write

$$W_p := \bigsqcup_{x \in \mathcal{C}_p} W_x, \quad V_p := \bigsqcup_{x \in \mathcal{C}_p} V_x.$$

Moreover, for a stable tangential structure $\xi : \mathcal{X} \rightarrow BO$, one can define $\mathcal{X}$ flow categories as those flow categories in which the morphism manifolds are equipped with an $\mathcal{X}$ structure that is compatible with composition (in place of a stable framing). For an $\mathcal{X}$ flow category $\mathcal{C}$, we obtain a coherent chain complex $K_\mathcal{C} \in \mathrm{Ch}(\mathrm{Mod}_{M\mathcal{X}})$, and, analogously to the realization in **Sp** (see [realization](#)), we define $|\mathcal{C}| \in \mathrm{Mod}_\mathcal{X}$ to be its iterated cofiber.

Our results

Let $\mathcal{X}$ be a stable tangential structure, let $\mathcal{C}$ be a $\mathcal{X}$ -structured flow category. An element $W_p = \{W_z\}_{|z|=p} \in E_1^{n,p}$ survives to the $(r+1)$ -th page if and only if it extends to a right flow module W of degree n over $\mathcal{C}_{[p,p-r]}$ (see [Definition 3.27](#) for the formal definition).

To describe the differential, let W, V be a right and left flow module over $\mathcal{C}$ of degree n and $p-r$ respectively (see [Definition 3.24](#), [Definition 3.25](#)). We define the punctured cube diagram

$$D : P(\{p, \dots, p-r\}) \setminus \{\emptyset\} \rightarrow \mathbf{Top},$$

where $P(-)$ denote the power set, by setting

$$D(\{a_k, \dots, a_1\}) = W_{a_k} \times \mathcal{M}_{a_k, a_{k-1}} \times \mathcal{M}_{a_{k-1}, a_{k-2}} \cdots \times \mathcal{M}_{a_2, a_1} \times V_{a_1},$$

where $a_k > a_{k-1} > \cdots > a_1$. The **Toda manifold** is the colimit which we denote

$$V \circ W := \mathrm{colim} D$$

and we will see later that the Toda manifolds indeed inherits manifold structure from V, W , and that they are closed, $(n-p+r)$ -dimensional, $\mathcal{X}$ -structured.

For the spectral sequence, we consider a family of such constructions indexed by objects in $\mathcal{C}_{p-(r+1)}$. Specifically, for each $z \in \mathcal{C}_{p-(r+1)}$, we define a punctured cube diagram D_z by replacing the left module V with $\mathcal{M}_{-,z}$ in the general construction. This yields a Toda manifold

$$\mathcal{M}_{-,z} \circ W := \text{colim } D_z \in \Omega_{n-p+r}^{\mathcal{X}}.$$

Taking the disjoint union over all $z \in \mathcal{C}_{p-(r+1)}$, we obtain an element

$$\bigsqcup_{z \in \mathcal{C}_{p-(r+1)}} (\mathcal{M}_{-,z} \circ W) \in \bigoplus_{z \in \mathcal{C}_{p-(r+1)}} \Omega_{n-p+r}^{\mathcal{X}}.$$

Example. In Fig. 2, 3, and 4, we can see an example of an extension and its Toda manifold.

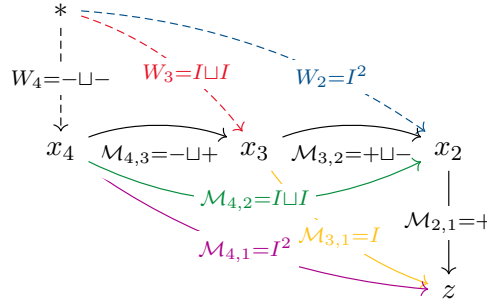


Figure 2: The solid part represents the flow category $\mathcal{C}$. Its objects are the x_i 's with $|x_i| = i$, and the arrows represent the morphism manifolds. The 0 dimensional framed manifold W_4 represents an element in $E_1^{4,4} = \Omega_{4-4}^{\text{fr}}$. The dashed arrows represent an right module over $\mathcal{C}_{[4,2]}$. The symbols $\pm$, I , etc., represent the flow manifolds themselves (where $\pm$ represents a framed point; the rest of framing is omitted for clarity).

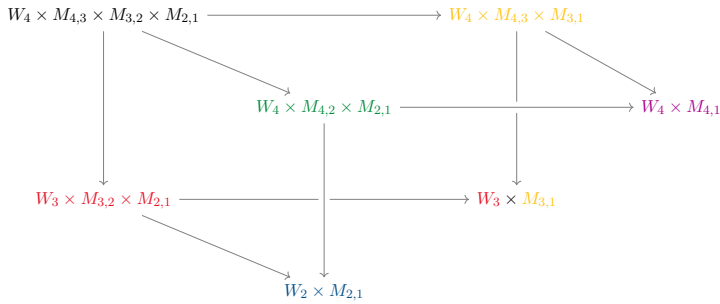


Figure 3: The punctured 3-cube diagram D_z associated with the object $z = x_1$ of degree 1, constructed from the right flow module W and the left flow module $\mathcal{M}_{-,z}$.

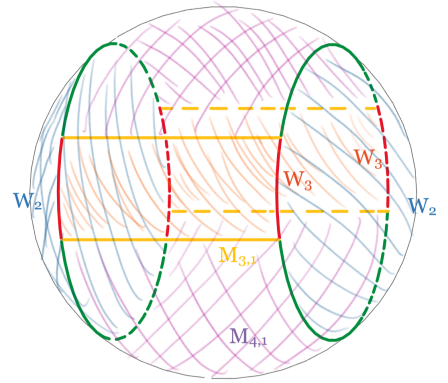


Figure 4: We can see that $\text{colim } D_z$ is a topological space that inherits a closed smooth manifold structure.

Theorem A. *Let $\mathcal{C}$ be a flow category with an $\mathcal{X}$ -tangential structure. There is a spectral sequence:*

$$E_1^{n,p} = \bigoplus_{|z|=p} \Omega_{n-p}^{\mathcal{X}}$$

which converges to the homotopy groups of the realization:

$$E_r^{n,p} \implies \pi_n |\mathcal{C}|.$$

An element

$$W_p = \{W_z\}_{|z|=p} \in E_1^{n,p}$$

survives to $E_{r+1}^{n,p}$ if and only if the collection W_p extends to a right flow module over $\mathcal{C}_{[p,p-r]}$ (see [Definition 3.27](#)). The differential is represented geometrically by the Toda manifolds:

$$d_{r+1}(W_p) = \bigcup_{y \in \mathcal{C}_{p-(r+1)}} (\mathcal{M}_{-,y} \circ W) \in \bigoplus_{y \in \mathcal{C}_{p-(r+1)}} \Omega_{n-p+r}^{\mathcal{X}} = E_{r+1}^{n-1,p-(r+1)}$$

A key tool used to prove [Theorem A](#) is a description of the spectral sequence associated with a coherent chain complex expressed in terms of chain complex maps.

Theorem B. *Let $K \in \text{Ch}(\mathbf{Sp})$ be a coherent chain complex in Spectra. Then there is a spectral sequence:*

$$E_1^{n,p} = \pi_{n-p} K_p,$$

which converges to

$$E_r^{n,p} \implies \pi_n \text{colim}_p (\mathcal{I}K)_p,$$

Moreover, let $x : \mathbb{S}^{n-p} \rightarrow K_p$ represent an element of $E_1^{n,p}$. Then x survives to $E_{r+1}^{n,p}$ if and only if there exists a chain complex map:

$$\begin{array}{ccccccc} \mathbb{S}^{n-p} & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 \\ \downarrow x & & \downarrow & & \downarrow & & \downarrow \\ K_p & \xrightarrow{d_p} & K_{p-1} & \xrightarrow{d_{p-1}} & \dots & \xrightarrow{d_{p-r}} & K_{p-r} \end{array} \quad (2)$$

Here, the top row is concentrated in degree p , and the bottom row is the restriction of K from degree p down to $p-r$. The differential $\partial_{r+1}(x)$ is given by the composition of chain maps:

$$\begin{array}{ccccc} \mathbb{S}^{n-p} & \longrightarrow & \dots & \longrightarrow & 0 \\ \downarrow x & & \downarrow & & \downarrow \\ K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r} \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \dots & \longrightarrow & K_{p-r-1} \end{array}$$

The first map is precisely [Eq. \(2\)](#) and the second is induced by the structure of the chain complex K .

We obtain this result by combining the description of the spectral sequence of a filtered object in Lurie [\[22\]](#) with the results of [\[5\]](#).

READ carefully about what conditions we need to get convergences

Relation to other work

This paper contributes to the broader project of providing a geometric model for spectra. For the foundation of this project, which defines the stable ∞ -category of framed flow categories and establishes its equivalence with the category of spectra, see [2].

This project traces back to the work of Cohen-Jones-Segal (CJS) in [10], where the term “flow category” was introduced as a means of organizing the data of a Morse-Smale function in a categorical way. In that work, a coherent version of chain complexes is constructed from such a category, and these chains are subsequently “realized” as a filtered spectrum. These ideas have since proven very useful in Floer theory.

In knot theory, organizing the data of knots into a flow category results in a stable homotopy type that lifts Khovanov homology (see [16]). Moreover, by simplifying the flow categories (see [20, 19]) and classifying 0- and 1-dimensional manifolds, one obtains a combinatorial description and an effective algorithm for computing the operations

$$\mathrm{Sq}^2 : H^*(\mathcal{C}, \mathbb{Z}/2) \rightarrow H^{*+2}(\mathcal{C}, \mathbb{Z}/2)$$

from a knot diagram. This leads to the calculation of the corresponding homotopy (see [21, 17]).

Our work is similar in spirit to these developments in that we use higher flow manifolds to compute the differentials, which can be interpreted as higher cohomology operations.

In Hamiltonian Floer theory, one of the breakthroughs related to the Arnold conjecture employed a CJS-like construction—a generalization of flow categories adapted to Hamiltonian flow data—to prove a version of the Arnold conjecture over a field of characteristic p (see [1]). Similar ideas appear in the work on the integral Arnold conjecture (see [33, 6]). Another recent development is the theory of Morse-Bott flow categories, where the function has critical sets (rather than isolated points) (see [8]). An ongoing line of research is the introduction of an equivariant structure on such categories, which in turn induces an equivariant structure on the realization spectrum—a framework that supports cyclotomic methods in symplectic geometry (see [11, 32]).

One also finds flow-category-like constructions in Lagrangian geometry that lift the Fukaya category of a symplectic manifold (see [30, 31]).

Remark 1.1. In this paper we treat flow categories as chain complexes in $\mathbf{Sp}$ or, equivalently, as filtered spectra. We expect that there is a construction analogous to that in [2] which provides a flow-categorical model for $\{K_p\} \in \mathrm{Ch}(\mathbf{Sp})$ such that $K_p = \bigoplus \mathbb{S}$.

The interpretation of gluing manifolds as Toda brackets or Massey products appears in [3], where the Massey product is defined via the gluing of two manifolds with boundary.

Theorem B bears a similar flavor to other works on spectral sequences associated with various objects. In [5], the authors explain the connection between filtered objects and coherent chain complexes, and they also define a spectral sequence for coherent chain complexes using the Beilinson t -structure. We also recommend [4] as an introduction to spectral sequences and their relation both to chain complexes and to the spectral sequence of filtered objects defined in [22]. The construction of the spectral sequence in this paper is similar to that in [7], where the authors define a spectral sequence associated with a simplicial object.

Paper layout

Section 2 is devoted to the framework of chain complexes in (stable) ∞ -categories and contains the proof of **Theorem B**.

Delete everything which is not directly related to flow categories (delete all the Khovanov, Arnold, knots,...).

Lagrangian Floer is general the the Hamiltonian so this wording needs to be revisited

delete this part about Arnold Con - maybe just mention that one can ask if our work can be generalized to Morse Bott flow categories / Equivariant flow categories

In Section 3, we recall the definitions of manifolds with corners, framed flow categories, and the Pontryagin-Thom construction for manifolds with corners. At the end of this section, we demonstrate several properties of the Pontryagin-Thom construction that will be useful in later sections.

Section 4 establishes a correspondence between chain complexes in the category of spectra (with values $\bigoplus \mathbb{S}$) and framed flow categories.

Our goal in Sections 5 and 6 is to upgrade this correspondence so that it interacts naturally with spectral sequences. In other words, given a framed flow category, we show how to produce a spectral sequence for the corresponding chain complex and provide a geometrical description of its pages and differentials.

In Section 5, we compare various models for chain complexes by considering different models for ∞ -categories (namely, topologically enriched, simplicially enriched, and simplicial sets). This comparison allows us to give a straightforward description of the composition of maps between flow categories.

Section 6 translates this description of composition into a geometrical framework and establishes the existence of a spectral sequence associated with framed flow categories.

In the final section, we explain how the framing structure can be replaced by a more general tangential structure, and we discuss how different structures interact. This section also contains examples of spectral sequences arising from flow categories.

Discussion and open problems

Different definitions for flow categories. In [2], the authors construct an ∞ -category Flow^{fr} (see also [30] for the homotopy category) and show that it is equivalent to $\mathbf{Sp}$. We now explain the main difference between their work and ours from an algebro-topological perspective. Let M be a compact manifold. Given two Morse-Smale functions f and g , one obtains two framed flow categories, $\mathcal{C}_f$ and $\mathcal{C}_g$. By interpolating between f and g , we obtain two maps in Flow^{fr}

$$H_{f,g} : \mathcal{C}_f \rightarrow \mathcal{C}_g, \quad H_{g,f} : \mathcal{C}_g \rightarrow \mathcal{C}_f,$$

and by homotoping the compositions we obtain 2-morphisms

$$H_{g,f} \circ H_{f,g} \simeq \text{Id}, \quad H_{f,g} \circ H_{g,f} \simeq \text{Id}.$$

Hence, the flow category of M is independent of the choice of Morse-Smale function.

On the other hand, the functions f and g induce CW structures on M , and different CW structures yield different (Atiyah-Hirzebruch) spectral sequences. Therefore, one cannot define a spectral sequence as an invariant of the equivalence classes in Flow^{fr} .

To address this issue, we conjecture that one can define another ∞ -category, with a modified class of morphisms. Specifically, for a 2-morphism R between flow categories $\mathcal{C}$ and $\mathcal{D}$, and for objects $x \in \mathcal{C}$ and $y \in \mathcal{D}$, we require that if $\|y\| > \|x\|$, then $R_{x,y} = \emptyset$ (rather than being a manifold of dimension $\|y\| - \|x\|$). This construction should yield a subcategory of $\text{Fil}(\mathbf{Sp})$ in which the two Morse-Smale functions on a manifold M produce distinct objects.

Using this perspective, we can think of [Theorem A](#) as a spectral sequence of filtered flow category $F \in \text{Fil}(\text{Flow}^{\text{fr}})$ where the filtration is induced by the index filtration on a flow category.

In this paper the full ∞ -categorical structure is unnecessary; instead, we define the equivalence relation on flow categories by pulling it back from the category $\text{Ch}(\mathbf{Sp})$ (see ??).

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2 Chains Complexes and their Spectral Sequences

Spectral sequences for filtered objects in Spectra are treated in Lurie’s work [22]. A key insight, developed by Ariotta and Krause [5], is that within a stable ∞ -category, filtered objects are equivalent to coherent chain complexes. Building upon this equivalence, this section aims to provide an explicit description of the spectral sequence associated with a coherent chain complex.

2.1 Chain Complexes and Filtered Objects

In this section, let $\mathcal{C}$ be a stable ∞ -category with sequential limits and colimits. Following [5], we will introduce the categories of chain complexes and filtered objects in ∞ -category. We will show how to connect between them and the equivalence between the two categories.

Remark 2.1. For being compatible with flow categories, our chain complexes will decrease and hence the filtered objects will increase, as oppose to [5].

Definition 2.2. We define the category $\mathbf{Ch}$ to be the pointed (ordinary) category with

$$\begin{aligned} \text{ob } \mathbf{Ch} &:= \mathbb{Z} \cup \{*\} \\ \text{Hom}_{\mathbf{Ch}}(i, j) &:= \begin{cases} \{\text{id}, \mathbf{0}\} & i = j \\ \{d_i, \mathbf{0}\} & i = j + 1 \\ \{\mathbf{0}\} & \text{otherwise} \end{cases} \end{aligned}$$

where $*$ is the zero object and $\mathbf{0}$ is the map $i \rightarrow * \rightarrow j$.

For $\mathcal{C}$ as above, we define *chain complexes in $\mathcal{C}$* (or just *chains*) to be the full subcategory

$$\mathbf{Ch}(\mathcal{C}) := \text{Fun}^*(\mathbf{Ch}, \mathcal{C}) \subset \text{Fun}(\mathbf{Ch}, \mathcal{C})$$

of pointed functors.

We denote by $\mathbf{Ch}_b$ the full subcategory of $\mathbf{Ch}(\mathcal{C})$ spanned by the objects K such that there exists $N \in \mathbb{N}$ with:

$$K_i = 0 \text{ for all } i < N.$$

We call such chain complexes *bounded below*.

Remark. If we don’t mention otherwise, we assume that $K \in \mathbf{Ch}_b$ is bounded below by $N = 0$.

Example. The category $\mathbf{Ch}(\mathbf{Ab})$ of chain complexes in abelian groups is the usual category of chain complexes.

Remark 2.3. In the case where $\mathcal{C}$ is an ∞ -category, the reader may want to think of $\mathbf{Ch}(\mathcal{C})$ as chain complexes, with null-homotopies of the composition of two maps along with higher null-homotopies: given a composition of three maps in the chain complex

$$K_p \rightarrow K_{p-1} \rightarrow K_{p-2} \rightarrow K_{p-3},$$

we get two null-homotopies of the map $K_p \rightarrow K_{p-3}$, which give us a map $S^1 \rightarrow \text{Hom}(K_p, K_{p-3})$. Part of the data of a chain complex is a null-homotopy of this loop and similarly higher null-homotopies for higher loops.

Definition 2.4. We denote by $\mathbb{Z}$ the category of integers with the usual increasing order.

Definition 2.5. Let $\mathcal{C}$ be a stable ∞ -category with sequential limits and colimits. We define the category of (increasing, complete) *filtered objects in $\mathcal{C}$* to be the full subcategory

$$\mathrm{Fil}(\mathcal{C}) \subset \mathrm{Fun}(\mathbb{Z}, \mathcal{C})$$

spanned by functors F such that $\lim_p F_p = 0$; sometimes denoted by $\widehat{\mathrm{Fil}}(\mathcal{C})$. We define the **bounded below filtered objects** to be the full subcategory

$$\mathrm{Fil}_b(\mathcal{C}) \subset \mathrm{Fil}(\mathcal{C})$$

spanned by functors F such that there exists $N \in \mathbb{N}$ with $F_p = 0$ for $p < N$.

Note that the image of the stable Yoneda embedding

$$\mathfrak{J}_{(-)} : \mathbb{Z}^{\mathrm{op}} \rightarrow \mathrm{Fun}(\mathbb{Z}, \mathbf{Sp}) = \mathrm{Fil}(\mathbf{Sp}).$$

is given by

$$\mathfrak{J}_p(q) = \mathrm{Map}_{\mathbf{Sp}}(p, q) = \begin{cases} \mathbb{S} & q \geq p \\ 0 & q < p \end{cases}$$

Let $p \rightarrow (p-1)$ be a morphism in $\mathbb{Z}^{\mathrm{op}}$, the morphisms $\mathfrak{J}_p \rightarrow \mathfrak{J}_{p-1}$ is given by

$$\begin{array}{ccccccc} \dots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{S} \longrightarrow \dots \\ & & \downarrow & & \downarrow & & \downarrow \\ \dots & \longrightarrow & 0 & \longrightarrow & \mathbb{S} & \longrightarrow & \mathbb{S} \longrightarrow \dots \end{array}$$

Definition 2.6. For $n, p \in \mathbb{Z}$ we define the following chain complex $\mathbb{S}_p^n \in \mathrm{Ch}(\mathbf{Sp})$ by

$$\mathbb{S}_p^n(q) := \begin{cases} \mathbb{S}^n & q = p \\ 0 & \text{otherwise} \end{cases}$$

Note that there is the map in $\mathrm{Ch}(\mathbf{Sp})$

$$Y_{n,p} : \mathbb{S}_p^{n-1} \rightarrow \mathbb{S}_{p-1}^n$$

which visually given by

$$\begin{array}{cccccccc} \dots & \longrightarrow & 0 & \longrightarrow & \mathbb{S}^{n-1} & \longrightarrow & 0 & \longrightarrow 0 \longrightarrow \dots \\ & & \downarrow & & \downarrow & & \downarrow & \\ \dots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{S}^n & \longrightarrow 0 \longrightarrow \dots \end{array}$$

which is defined using the fact that $\mathbb{S}^n \simeq \Sigma \mathbb{S}^{n-1}$ in $\mathbf{Sp}$.

In particular we have a map

$$Y_p := Y_{-(p-1),p} : \mathbb{S}_p^{-p} \rightarrow \mathbb{S}_{p-1}^{-(p-1)}$$

Definition 2.7. We define

$$\mathcal{A}: \text{Fil}(\mathbf{Sp}) \rightarrow \text{Ch}(\mathbf{Sp})$$

to be the colimit preserving functor that determined by mapping

$$\mathcal{A}(\mathfrak{J}_p) := \mathbb{S}_p^{-p}$$

and the morphisms are given by Y_p .

We denote it's right adjoint

$$\text{Fil}(\mathbf{Sp}) \leftarrow \text{Ch}(\mathbf{Sp}) : \mathcal{I}$$

In [5], Ariotta define these functors and proves

Theorem 2.8 ([5]). *Let $\mathcal{C}$ be a stable ∞ -category with sequential limits and colimits. We have an equivalence*

$$\mathcal{A} : \text{Fil}(\mathcal{C}) \simeq \text{Ch}(\mathcal{C}) : \mathcal{I}.$$

An immediate consequence is the representability of the evaluation:

$$\text{Map}(\mathbb{S}_p^{-p}, \mathcal{A}F) \simeq \text{Map}(\mathbb{I}\mathbb{S}_p^{-p}, F) \simeq \text{Map}(\mathfrak{J}_p, F) \simeq F_p \quad (3)$$

Definition 2.9. Let $\mathcal{C}$ be a stable ∞ -category and $p \in \mathbb{Z}$, we write $\iota_{[p-1, p]} : \{(p-1) \rightarrow p\} \rightarrow \mathbb{Z}$ for the inclusion functor and

$$(\iota_{[p-1, p]})^* : \text{Fun}(\mathbb{Z}, \mathcal{C}) \rightarrow \text{Fun}(\Delta^1, \mathcal{C})$$

for the restriction (pre-composition) along $\iota_{[p-1, p]}$. We define the p -th associated graded functor by:

$$\text{gr}_p := \text{cof} \circ (\iota_{[p-1, p]})^* : \text{Fun}(\mathbb{Z}, \mathcal{C}) \rightarrow \mathcal{C},$$

so that

$$F \mapsto F_p / F_{p-1}.$$

The map

$$F_p / F_{p-1} \rightarrow \Sigma F_{p-1} / F_{p-2}$$

given by the connecting morphism of the cofiber, and then taking quotient, induces a transformation:

$$\text{gr}_p \rightarrow \Sigma \text{gr}_{p-1}.$$

A direct calculation shows that

$$\Sigma^{-p} \text{gr}_q(\mathfrak{J}_p) = (\mathcal{A}\mathfrak{J}_p)_q$$

and we deduce that

Corollary 2.10.

$$(\mathcal{A}-)_p \simeq \Sigma^{-p} \text{gr}_p(-)$$

As functors

$$\text{Fil}(\mathbf{Sp}) \rightarrow \mathbf{Sp}$$

We now provide an explicit expression for $\mathcal{I}$.

First, we define a the following chain complex in $\mathbf{Sp}$,

$$\Sigma^n y_p := \begin{cases} \mathbb{S}^n & q \in \{p, p-1\} \\ 0 & \text{otherwise} \end{cases}$$

In particular we have

$$\mathrm{cof}(Y_p) \simeq \Sigma^{-(p-1)} y_p$$

On the other hand (see [5] Lemma 3.12), one can show that the evaluation functor

$$(-)_p: \mathrm{Ch}(\mathbf{Sp}) \rightarrow \mathbf{Sp}, \quad K \mapsto K_p$$

is represented:

$$\mathrm{Map}(y_p, -) \simeq (-)_p.$$

The objects in the following fiber sequence

$$\mathbb{S}_p^{-p} \rightarrow \mathbb{S}_{p-1}^{-(p-1)} \rightarrow \mathrm{cof}(Y_p) \tag{4}$$

represents the functors

$$(\mathcal{I}-)_p, \quad (\mathcal{I}-)_{p-1}, \quad \Sigma^{-(p-1)}(-)_p.$$

By taking the contravariant Map functor to Eq. (4) we get the following result:

Lemma 2.11. *There is a cofiber sequence of functors*

$$\Sigma^{-(p-1)}(-)_p \rightarrow (\mathcal{I}-)_{p-1} \rightarrow (\mathcal{I}-)_p \tag{5}$$

A simple corollary is

Corollary 2.12. *Let $K \in \mathrm{Ch}_b(\mathbf{Sp})$ bounded below by N . Then for $p \leq N$ we have*

$$(\mathcal{I}K)_p = 0$$

which obtained by inductively applying Lemma 2.11.

Example. Let

$$\cdots \rightarrow 0 \rightarrow F_0 \rightarrow F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow 0 \rightarrow \cdots$$

be a filtered object in $\mathbf{Sp}$. Then $K := \mathcal{A}F$ is given by:

$$\cdots \rightarrow 0 \rightarrow \Sigma^{-3}F_3/F_2 \rightarrow \Sigma^{-2}F_2/F_1 \rightarrow \Sigma^{-1}F_1/F_0 \rightarrow F_0 \rightarrow 0 \cdots$$

where the zeros on the left are F_3/F_3 . The composition

$$K_3 = \Sigma^{-3}F_3/F_2 \rightarrow \Sigma^{-2}F_2/F_1 \rightarrow \Sigma^{-1}F_1/F_0 = K_1$$

can be factored as:

$$\Sigma^{-3}F_3/F_2 \rightarrow \Sigma^{-2}F_2 \rightarrow \Sigma^{-2}F_2/F_1 \rightarrow \Sigma^{-1}F_1 \rightarrow \Sigma^{-1}F_1/F_0,$$

where the middle part is a cofiber sequence; hence, the composition is null-homotopic. Following [Remark 2.3](#), this null-homotopy is the "non-higher" part of the chain complex K .

To recover $F = \mathcal{I}K$ from K , we can use [Lemma 2.11](#). The fact that K is bounded below by -1 yields that

$$(\mathcal{I}K)_p = 0 \quad \text{for } p \leq -1$$

From that following cofiber sequence We can see that $F_0 = K_0$

$$\Sigma^{-(-1)}K_0 \rightarrow \overbrace{F_{-1}}^{=0} \rightarrow F_0.$$

We can recover F_1 as the cofiber

$$K_1 \rightarrow \overbrace{F_0}^{=K_0} \rightarrow F_1.$$

And by taking F_2 is reconstructed as an iterated cofiber, from the cofiber sequence :

$$\Sigma^{-1}K_2 \rightarrow \overbrace{F_1}^{=K_1/K_0} \rightarrow F_2.$$

Remark 2.13. An "non example" is given by taking a topological space X and the filtered space $F_p := \text{sk}_p X$. If X is finite, then there is an $N \in \mathbb{N}$ - an upper bound for the filtration. In this case, we can define the following chain complex in **Spaces**

$$K_p := \Sigma^{N-p} \text{gr}_p F = \text{sk}_p X / \text{sk}_{p-1} X \simeq \bigvee_{p \text{ cells}} S^N$$

and the morphisms $\text{gr}_p \rightarrow \Sigma \text{gr}_{p-1}$ that induces the maps $K_p \rightarrow K_{p-1}$, are given by the maps:

$$\bigvee S^p = \text{sk}_p X / \text{sk}_{p-1} X \xrightarrow{\partial_p} \text{sk}_{p-1} X / \text{sk}_{p-2} X = \Sigma \bigvee S^{p-1},$$

which are more commonly known as the differentials in the chain complex that computes the cellular homology of X . By taking iterated cofibers we can recover F only up to suspension, for example

$$\Sigma^N F_1 \simeq \text{cof}(\Sigma^{N-1} F_1 / F_0 \rightarrow S^N F_0)$$

2.2 Spectral Sequences of Chain Complexes

We will start with the construction of spectral sequence of a filtered object, following the notations and definitions in [\[23\]](#). Combined with the last subsection we will get a description of the pages of a spectral sequence of a chain complex. The main output is [Proposition 2.24](#).

For our discussion, we assume that $\mathcal{C}$ is a stable ∞ category equipped with a t -structure $(\mathcal{C}^{\leq 0}, \mathcal{C}^{\geq 0})$. We begin by defining the spectral sequence associated with a filtered object in $\mathcal{C}$.

Definition 2.14. [\[22, Def. 1.2.2.9\]](#) Let F be a filtered object in $\mathcal{C}$, i.e., a sequence of morphisms $\{F_p\}_{p \in \mathbb{Z}}$ such that

$$\cdots \rightarrow F_{p-1} \rightarrow F_p \rightarrow F_{p+1} \rightarrow \cdots. \quad (6)$$

For each $p \in \mathbb{Z}$, define the graded pieces by

$$\text{gr}_p F = \text{cof}(F_{p-1} \rightarrow F_p). \quad (7)$$

We define a spectral sequence $\{E_r^{n,p}, d_r^{n,p}\}$ associated with F by setting

$$E_r^{n,p} = \text{Im}(\pi_n(F_p/F_{p-r}) \rightarrow \pi_n(F_{p+r-1}/F_{p-1})), \quad (8)$$

where the map is induced by the compositions of the inclusion and projection:

$$F_p/F_{p-r} \rightarrow F_p/F_{p-1} \rightarrow F_{p+r-1}/F_{p-1}. \quad (9)$$

The differentials $d_r^{n,p}: E_r^{n,p} \rightarrow E_r^{n-1,p-r}$ are determined by the connecting morphisms via the commutativity of the diagram:

$$\begin{array}{ccccc} \pi_n(F_p/F_{p-r}) & \longrightarrow & E_r^{n,p} & \longrightarrow & \pi_n(F_{p+r-1}/F_{p-1}) \\ \downarrow \delta_{p-r} & & \downarrow d_r^{n,p} & & \downarrow \delta_{p-1} \\ \pi_{n-1}(F_{p-r}/F_{p-2r}) & \longrightarrow & E_r^{n-1,p-r} & \longrightarrow & \pi_{n-1}(F_{p-1}/F_{p-r-1}), \end{array} \quad (10)$$

where the vertical maps δ_{p-r} and δ_{p-1} are the connecting homomorphisms from the long exact sequences of homotopy groups associated with the cofiber sequences:

$$F_{p-r} \rightarrow F_p \rightarrow F_p/F_{p-r} \quad \text{and} \quad F_{p-1} \rightarrow F_{p+r-1} \rightarrow F_{p+r-1}/F_{p-1}. \quad (11)$$

Theorem 2.15. [22, Prop. 1.2.2.14] *Let $\mathcal{C}$ be a stable ∞ -category equipped with a t -structure, that admits sequential colimits and that the t -structure is compatible with sequential colimits. Let F be a complete filtered object in $\mathcal{C}$. Then the spectral sequence $\{E_r^{n,p}, d_r^{n,p}\}$ (see Definition 2.14) converges to the homotopy groups of the colimit of F :*

$$E_r^{n,p} \Rightarrow \pi_n \text{colim}_p F_p.$$

Example 2.16. Consider a CW complex X with its skeletal filtration $\{\text{sk}_p X\}_{p \geq 0}$. Applying the suspension spectrum functor Σ^∞ , we obtain a filtered spectrum F in $\mathbf{Sp}$, where $F_p = \Sigma^\infty \text{sk}_p X$. Let $K = \mathcal{A}F$ be the associated chain complex (as in Remark 2.13). The spectral sequence arising from this filtration has first page:

$$E_1^{n,p} = \pi_n \text{gr}_p F = \pi_n \left(\bigoplus \mathbb{S}^p \right) = \pi_{n-p} K_p,$$

since each $\text{gr}_p F$ is a wedge of spheres suspended p times, and K_p is the p -th term in the chain complex K .

This spectral sequence converges to:

$$\pi_n \Sigma^\infty X,$$

which are the stable homotopy groups of the space X .

More generally, for any spectrum A , we can consider the filtered spectrum $F_p \otimes A$, obtaining a spectral sequence converging to the generalized homology:

$$A_n(X).$$

This spectral sequence is known as the *Atiyah–Hirzebruch spectral sequence*.

Definition 2.17. Let F be a filtered object in stable ∞ -category $\mathcal{C}$ and let $E_r^{n,p}$ be it's spectral sequence. An element $x \in E_1^{n,p}$ is said to **survive to the r -th page** if it lies in the image of the map:

$$x \in \text{Im}(\pi_n(F_p/F_{p-r}) \rightarrow \pi_n(F_p/F_{p-1})).$$

We denote the collection of such elements by $\mathcal{E}_r^{n,p}$. If $x' \in \pi_n(F_p/F_{p-r})$ maps to x , we say that x' **lifts** x .

Intuitively, an element survives to the r -th page if it is not killed by the differentials d_k for $1 \leq k < r$.

Remark 2.18. In the stable homotopy category $\mathbf{Sp}$, the homotopy groups π_n are representable by spheres $\mathbb{S}^n$. Thus, elements in

$$E_1^{n,p} = \pi_n \text{gr}_p F = \pi_{n-p} K_p$$

are represented by maps $x: \mathbb{S}^{n-p} \rightarrow K_p$.

We introduce a notation for subcategories of chain complexes.

Definition 2.19. For integers $a \leq b$, let $\text{Ch}_{[b,a]}$ denote the full subcategory of the poset category Ch spanned by the objects

$$\{i \in \mathbb{Z} \mid b \geq i \geq a\} \cup \{*\}, \quad (12)$$

where $*$ denotes the initial object. Given a pointed category $\mathcal{C}$, we define

$$\text{Ch}_{[b,a]}(\mathcal{C}) = \text{Fun}^*(\text{Ch}_{[b,a]}, \mathcal{C}) \subseteq \text{Fun}(\text{Ch}_{[b,a]}, \mathcal{C}), \quad (13)$$

the full subcategory of pointed functors from $\text{Ch}_{[b,a]}$ to $\mathcal{C}$.

Precomposition with the inclusion $\text{Ch}_{[b,a]} \hookrightarrow \text{Ch}$ induces restriction functors:

$$i_{[b,a]}^*: \text{Ch}(\mathcal{C}) \rightarrow \text{Ch}_{[b,a]}(\mathcal{C}).$$

Given $K \in \text{Ch}(\mathcal{C})$, its restriction to $\text{Ch}_{[b,a]}$ is depicted as:

$$K_b \rightarrow K_{b-1} \rightarrow \cdots \rightarrow K_a.$$

Restrictions from below admits the following right adjoint:

$$(i_{\geq a})_*: \text{Ch}_{\geq a}(\mathbf{Sp}) \rightarrow \text{Ch}(\mathbf{Sp})$$

which given by padding with zeros:

$$(i_{\geq a})_* K(p) = \begin{cases} K(p) & p \geq a \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

On the other hand, restriction from above have padding as left adjoint:

$$(i_{\leq b})! K(p) = \begin{cases} 0 & p > b \\ K(p) & b \geq p \end{cases} \quad (15)$$

We denote the composition by:

$$K \mapsto K_{[b,a]} =: (i_{\leq b})!(i_{\geq a})_* K$$

We can do the same for filtered object.

Definition 2.20. For two integers $a \leq b$ we define

$$\mathrm{Fil}_{[a,b]}(\mathbf{Sp}) := \mathrm{Fun}([a,b], \mathbf{Sp})$$

Note that the inclusion

$$i_{\leq n} : (-\infty, n] \rightarrow \mathbb{Z}$$

induces the restriction functor:

$$i_{\leq n}^* : \mathrm{Fil}(\mathbf{Sp}) \rightarrow \mathrm{Fil}_{(-\infty, n]}(\mathbf{Sp}) =: \mathrm{Fil}_{\leq n}(\mathbf{Sp})$$

and left adjoint

$$(i_{\leq n})_! : \mathrm{Fil}_{\leq n}(\mathbf{Sp}) \rightarrow \mathrm{Fil}(\mathbf{Sp})$$

which is given by

$$(i_{\leq n})_! F(p) = \begin{cases} F(p) & p \leq n \\ F(n) & \text{otherwise} \end{cases}$$

Similarly we define

$$\mathrm{Fil}_{\geq n}(\mathbf{Sp})$$

The map

$$i_{\geq n} : [n, \infty) \rightarrow \mathbb{Z}$$

induces the restriction

$$i_{\geq n}^* : \mathrm{Fil}(\mathbf{Sp}) \rightarrow \mathrm{Fil}_{\geq n}(\mathbf{Sp})$$

and it's right adjoint

$$(i_{\geq n})_* F(p) = \begin{cases} F(p) & n \leq p \\ 0 & \text{otherwise} \end{cases}$$

Note. sometimes we will identify between $F \in \mathrm{Fil}_{[n,m]}(\mathbf{Sp})$ to it's image in $\mathrm{Fil}(\mathbf{Sp})$.

On the other hand we can define the "quotient truncation" functor.

Definition 2.21. Let $n \in \mathbb{Z}$ we define

$$q_n : \mathrm{Fil}(\mathbf{Sp}) \rightarrow \mathrm{Fil}_{\geq n}(\mathbf{Sp})$$

by

$$(q_n F)_p := \begin{cases} F_p / F_n & n+1 \leq p \\ 0 & \text{otherwise} \end{cases}$$

This functor behave nicely with the equivalence to chain complexes:

Corollary 2.22. Let $F \in \mathrm{Fil}(\mathbf{Sp})$, we have that:

$$\mathcal{A}q_n F \simeq i_{\geq n+1}^* \mathcal{A}F$$

treated as objects in $\mathrm{Ch}_{[\infty, n+1]}(\mathbf{Sp})$ or $\mathrm{Ch}(\mathbf{Sp})$.

Proof. A direct calculation shows that both are given by $(-)_p = \Sigma^{-p} \mathrm{gr}_p F$. □

Lemma 2.23. *Let $K \in \text{Ch}(\mathbf{Sp})$ and $F = \mathcal{I}K$ its equivalent filtered spectrum, then*

$$\text{Map}(\mathbb{S}_p^{-p}, i_{\geq n+1}^* K) \simeq F_p/F_n$$

Proof. Note that $\mathcal{A}F = K$. We use the corollary above and Eq. (3) to get:

$$\text{Map}(\mathbb{S}_p^{-p}, i_{\geq n+1}^* K) \simeq \text{Map}(\mathbb{S}_p^{-p}, \mathcal{A}q_n F) \simeq (q_n F)_p = F_p/F_n.$$

□

Let K be a chain complex, and let $F = \mathcal{I}K$, then by passing Lurie's construction of spectral sequence of F we get the following

Proposition 2.24. *Let $K \in \text{Ch}(\mathbf{Sp})$ be a chain complex in Spectra, let $F = \mathcal{I}K$ the equivalent filtered spectrum and $E_r^{n,p}$ it's spectral sequence (see Definition 2.14 and Theorem 2.15). By passing Lurie's construction we get that*

$$E_1^{n,p} = \pi_{n-p} K_p,$$

Moreover, let $x: \mathbb{S}^{n-p} \rightarrow K_p$ represent an element in $E_1^{n,p}$. Then x survives to E_r if and only if there are maps of chain complexes:

$$\begin{array}{ccccccc} \mathbb{S}^{n-p} & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 \\ x \downarrow & & \downarrow & & \downarrow & & \downarrow \\ K_p & \xrightarrow{d_p} & K_{p-1} & \xrightarrow{d_{p-1}} & \dots & \xrightarrow{d_{p-r+2}} & K_{p-r+1}, \end{array}$$

where the top row is concentrated in degree p , and the bottom row is the truncated chain complex from degree p to $p-r+1$.

Proof. The first page is given in Remark 2.18. Similar to $i_{\geq n}$, the inclusion

$$j := i_{\geq p-1, p-r}: (\infty, p-1] \rightarrow (\infty, p-r]$$

induces the restriction functor j^* . By post composing we get a map

$$\text{Map}(\mathbb{S}_p^{-p}, i_{\geq p-r+1}^* K) \xrightarrow{j^* \circ} \text{Map}(\mathbb{S}_p^{-p}, i_{\geq p}^* K).$$

By suspending both sides (from $\mathbb{S}_p^{-p}$ to $\mathbb{S}_p^{n-p}$) we get the map

$$\text{Map}(\mathbb{S}_p^{n-p}, i_{\geq p-r+1}^* K) \xrightarrow{j^* \circ} \text{Map}(\mathbb{S}_p^{n-p}, i_{\geq p}^* K). \quad (16)$$

To get that visualization in the proposition, we first clarify that we use $\mathbb{S}_p^{n-p}$ for both the restricted and the usual versions of the chain that concentrated at p and we can write:

$$\begin{aligned} & \text{Map}_{[p, p-r+1]}(\mathbb{S}_p^{n-p}, K_{[p, p-r+1]}) \\ &= \text{Map}_{[p, p-r+1]}(\mathbb{S}_p^{n-p}, i_{\leq p}^* i_{\geq p-r+1}^* K) \simeq \text{Map}_{(\infty, p-r+1]}((i_{\leq p})! \mathbb{S}_p^{n-p}, i_{\geq p-r+1}^* K) \\ &= \text{Map}_{(\infty, p-r+1]}(\mathbb{S}_p^{n-p}, i_{\geq p-r+1}^* K) \simeq \Sigma^{n-p} F_p/F_{p-r} \end{aligned}$$

where the equalities are notations and the last equivalence is [Lemma 2.23](#). We can do the same to the right hand side of [Eq. \(17\)](#) and get the map:

$$\Sigma^{-n}F_p/F_{p-r} \simeq \text{Map}(\mathbb{S}_p^{n-p}, K_{[p, p-r+1]}) \xrightarrow{j^* \circ} \text{Map}(\mathbb{S}_p^{n-p}, K_p) \simeq \Sigma^{-n}F_p/F_{p-1} \quad (17)$$

By taking π_0 we conclude that $x \in \pi_n F_p/F_{p-1}$ is in the image of $\pi_n F_p/F_{p-r}$ if and only if the representing map

$$x : \mathbb{S}^n \rightarrow K_p$$

factor through $K_{[p, p-r+1]}$, which conclude the proof. $\square$

Example 2.25. Returning to [Example 2.16](#), consider an element $x : \mathbb{S}^{n-p} \rightarrow K_p$ representing a class in $E_1^{n,p}$. The differential $d_1^{n,p} : E_1^{n,p} \rightarrow E_1^{n-1, p-1}$ is given by:

$$d_1^{n,p}(x) = d_p \circ x, \quad (18)$$

where $d_p : K_p \rightarrow K_{p-1}$ is the differential in the chain complex K .

By [Proposition 2.24](#), x survives to the second page if and only if $d_p \circ x$ is null-homotopic, i.e., x is a cycle:

$$d_p \circ x = 0 \quad \text{in} \quad \pi_{n-p} K_{p-1}. \quad (19)$$

Similarly, x survives to the r -th page if it extends to a map of chain complexes over r steps, corresponding to being a cycle up to r -th order.

2.3 The differential from chain complexes viewpoint

In the previous subsections, we established the construction of a spectral sequence associated with a filtered object in a stable ∞ -category $\mathcal{C}$ equipped with a t -structure. We now focus on describing the differentials in this spectral sequence, particularly in the context of chain complexes. Let F be a filtered object in $\mathcal{C}$. Recall from [Definition 2.14](#) that we have the spectral sequence $\{E_r^{n,p}, d_r^{n,p}\}$ associated with F . To understand the differentials $d_r^{n,p}$, we consider the following commutative diagram arising from the connecting morphisms in the long exact sequences of homotopy groups:

$$\begin{array}{ccccc} \pi_n(F_p/F_{p-r}) & \longrightarrow & \pi_n(F_p/F_{p-1}) & \longrightarrow & \pi_n(F_{p+r-1}/F_{p-1}) \\ \delta_{p-r} \downarrow & & & & \downarrow \delta_{p-1} \\ \pi_{n-1}(F_{p-r}/F_{p-2r}) & \longrightarrow & \pi_{n-1}(F_{p-r}/F_{p-r-1}) & \longrightarrow & \pi_{n-1}(F_{p-1}/F_{p-r-1}), \end{array} \quad (20)$$

where the vertical maps δ_{p-r} and δ_{p-1} are the connecting homomorphisms from the long exact sequences associated with the cofiber sequences in the filtration.

Let $x \in E_r^{n,p}$ be an element that survives to the r -th page, and let $x' \in \pi_n(F_p/F_{p-r})$ be a lift of x . By the commutativity of [Diagram Equation \(20\)](#), the image of $\delta_{p-r}(x')$ in $\pi_{n-1}(F_{p-1}/F_{p-r-1})$ depends only on x and not on the specific choice of the lift x' . This property allows us to define the differential $d_r^{n,p}$.

Definition 2.26. Let F be a filtered spectrum with spectral sequence $E_r^{n,p}$ and let $x \in E_r^{n,p}$. Let $x' \in \pi_n(F_p/F_{p-r})$ be a lift of x . We define the differential

$$d_r^{n,p}(x) \in E_r^{n-1, p-r}$$

Combine
with sec
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to be the image of $\delta_{p-r}(x')$ in $E_r^{n-1,p-r}$ under the natural map induced by the composition

$$\pi_{n-1}(F_{p-r}/F_{p-2r}) \rightarrow E_r^{n-1,p-r}.$$

Diagrammatically, this is represented as:

$$\begin{array}{ccc} x' & \xrightarrow{\quad} & x \\ \delta_{p-r} \downarrow & & \\ \delta_{p-r}(x') & \longmapsto & d_r^{n,p}(x). \end{array} \quad (21)$$

The definition ensures that $d_r^{n,p}(x)$ is well-defined, independent of the choice of lift x' .

Chain-level description of the differential

Similar to the description of the pages, as maps between chain complexes, we can unwind Lurie's description for filtered spectrum, to get a chain complex description of the differential.

To facilitate the proof of [Proposition 2.29](#), we first introduce the following definition and lemma.

Definition 2.27. We denote the "shift to the left" functor by

$$T: \text{Ch}(\mathbf{Sp}) \rightarrow \text{Ch}(\mathbf{Sp})$$

which is given by precomposing with

$$-1: \mathbb{Z} \rightarrow \mathbb{Z}, \quad n \mapsto n - 1$$

Lemma 2.28. *Let $L \in \text{Ch}_{(\infty,n]}(\mathbf{Sp})$ and $K \in \text{Ch}_{[n,-\infty)}(\mathbf{Sp})$ be two chain complexes. Then*

$$\text{Map}(\Sigma^{-1}TL, K) \simeq \text{Map}(L, K)$$

Proof. First, note that $\text{Fil}(\mathbf{Sp})$ is generated (under colimits) by $\mathfrak{J}_p$. Passing to $\text{Ch}(\mathbf{Sp})$, we get that $\text{Ch}(\mathbf{Sp})$ is generated by $\mathcal{I}\mathfrak{J}_p = \mathbb{S}_p^{-p}$. Using this observation, it suffices to show that:

$$\text{Map}(\mathbb{S}_p^{-p}, K) \simeq \text{Map}(\Sigma^{-1}T\mathbb{S}_p^{-p}, K) \simeq \text{Map}(S_{p+1}^{-(p+1)}, K)$$

and by taking $\mathcal{I}$, our goal is to show:

$$\text{Map}(\mathfrak{J}_p, \mathcal{I}K) \simeq \text{Map}(\mathfrak{J}_{p+1}, \mathcal{I}K)$$

which is equivalent to

$$(\mathcal{I}K)_p \simeq (\mathcal{I}K)_{p+1} \quad (22)$$

Note that the assumption on L translates to $p \geq n$.

Let $F = \mathcal{I}K$, where $K \in \text{Ch}_{[n,-\infty)}(\mathbf{Sp})$. Then for $q \geq n + 1$ we have

$$0 = K_q = (\mathcal{A}F)_q = \Sigma^{-q}F_q/F_{q-1}$$

which shows that [Eq. \(22\)](#) holds. □

Proposition 2.29. *Let $K \in \text{Ch}(\mathbf{Sp})$ be a chain complex in Spectra, let $F = \mathcal{I}K$ be its associated filtered spectrum, and let $E_r^{n,p}$ be the spectral sequence of F . Let $x \in \mathcal{E}_r^{n,p}$ be an element surviving to the r -th page (see [Definition 2.17](#)). Then the differential $\partial_r(x)$ is given by the composition of chain maps:*

$$\begin{array}{ccccc}
 \mathbb{S}^{n-p} & \longrightarrow & \dots & \longrightarrow & 0 \\
 \downarrow x & & \downarrow & & \downarrow \\
 K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r+1} \\
 \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \dots & \longrightarrow & K_{p-r}
 \end{array}$$

viewed as a map $\mathbb{S}^{n-p} \rightarrow K_{p-r}$, where the first map is an extension of x (see [Proposition 2.24](#)), and the second is induced by the structure of the chain complex K .

Proof. We begin with Lurie's description, [Eq. \(10\)](#). The differential is given by the map:

$$\pi_n F_p / F_{p-r} \rightarrow \pi_{n-1} F_{p-r} / F_{p-2r} \quad (23)$$

hence ∂_r is induced by post-composing with the connecting morphism:

$$F_p / F_{p-r} \rightarrow \Sigma F_{p-r} / F_{p-2r}$$

which comes from the cofiber sequence:

$$F_{p-r} / F_{p-2r} \rightarrow F_p / F_{p-2r} \rightarrow F_p / F_{p-r}.$$

As we saw in the proof of [Proposition 2.24](#), we can use [Lemma 2.23](#) to see that the map $F_p / F_{p-2r} \rightarrow F_p / F_{p-r}$ is representable as

$$j^*: \text{Map}(\mathbb{S}_p^{-p}, K_{[p, p-2r+1]}) \xrightarrow{j^\circ} \text{Map}(\mathbb{S}_p^{-p}, K_{[p, p-r+1]})$$

which is given by postcomposing with restriction and padding. By taking the cofiber of j , we get the cofiber of j^* . The cofiber is calculated pointwise: for $q \in [p-r, p-2r+1]$, the map j^* is $K_q \rightarrow 0$, and for $q \notin [p-r, p-2r+1]$, the map is the identity. We conclude that the cofiber map is of the form:

$$c: K_{[p, p-r+1]} \rightarrow \text{cof}(j^*) \simeq \Sigma K_{[p-r, p-2r+1]}.$$

Hence, the differential is represented by post-composing:

$$c^*: \text{Map}(\mathbb{S}_p^{-p}, K_{[p, p-r+1]}) \xrightarrow{c^\circ} \text{Map}(\mathbb{S}_p^{-p}, \Sigma K_{[p-r, p-2r+1]}).$$

Note that $\mathbb{S}_p^{-p} \in \text{Ch}_{(\infty, p]}(\mathbf{Sp})$ and the other two chains are elements in $\text{Ch}_{[p, -\infty)}(\mathbf{Sp})$. So we can use [Lemma 2.28](#) on c and find that c^* is equivalent to post-composition with

$$K_{[p, p-r+1]} \rightarrow T^{-1} K_{[p-r, p-2r+1]}$$

which is the visualization in the proposition. Note that the diagram (which has K_{p-r} as a target) is a visualization of $\partial_r(x) \in E_1^{n-1, p-r}$, and the "non-truncated" version (with target $K_{[p-r, p-2r+1]}$) is the evidence that we can lift $\partial_r(x)$ to $\mathcal{E}_r^{n-1, p-r}$.

Let $K = \mathcal{A}F$, we get a cofiber sequence of chain maps:

$$\begin{array}{ccccccccc}
 0 & \longrightarrow & \dots & \longrightarrow & 0 & \longrightarrow & K_{p-r} & \longrightarrow & \dots & \longrightarrow & K_{p-2r+1} \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r+1} & \longrightarrow & K_{p-r} & \longrightarrow & \dots & \longrightarrow & K_{p-2r+1} \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r+1} & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0
 \end{array}$$

Colimits in the functor category of chains are computed pointwise, hence the connecting morphism is given by

$$\begin{array}{ccccccccc}
 K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r+1} & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \dots & \longrightarrow & 0 & \longrightarrow & \Sigma K_{p-r} & \longrightarrow & \dots & \longrightarrow & \Sigma K_{p-2r+1}
 \end{array} \tag{24}$$

We denote the restriction of chains:

$$\begin{aligned}
 K_{[p,p-r+1]} &:= K_p \rightarrow \dots \rightarrow K_{p-r+1} \rightarrow \overbrace{0 \rightarrow \dots \rightarrow 0}^{r-1} \\
 K_{[p-r,p-2r+1]} &:= \overbrace{0 \rightarrow \dots \rightarrow 0}^{r-1} \rightarrow K_{p-r} \rightarrow \dots \rightarrow K_{p-2r+1}
 \end{aligned}$$

as objects in $\text{Ch}_{[p,p-2r+2]}(\mathbf{Sp})$. The diagram Eq. (24) is equivalent to the diagram:

$$\begin{array}{ccc}
 K_{[p,p-r+1]} & \longrightarrow & 0 \\
 \downarrow & & \downarrow \\
 0 & \longrightarrow & \Sigma K_{[p-r,p-2r+1]}
 \end{array}$$

Using the stability of $\mathbf{Sp}$, we get that $\text{Ch}_{[p,p-2r+2]}(\mathbf{Sp})$ is stable, and the last diagram is equivalent to the map of chains: $K_{[p,p-r+1]} \rightarrow K_{[p-r,p-2r+1]}$ which can be drawn as:

$$\begin{array}{ccccccc}
 K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r+1} & \longrightarrow & \dots & \longrightarrow & 0 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \dots & \longrightarrow & K_{p-r} & \longrightarrow & \dots & \longrightarrow & K_{p-2r+1}
 \end{array}$$

Hence, pre-composition with a lift of x to a map $x' : \mathbb{S}_p^{n-p} \rightarrow K_{[p,p-r+1]}$ gives the diagram:

$$\begin{array}{ccccccc}
 \mathbb{S}_p^{n-p} & \longrightarrow & \dots & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 \\
 \downarrow x & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r+1} & \longrightarrow & \dots & \longrightarrow & 0 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \dots & \longrightarrow & K_{p-r} & \longrightarrow & \dots & \longrightarrow & K_{p-2r+1}
 \end{array} \tag{25}$$

and the truncated version:

$$\begin{array}{ccccc}
 \mathbb{S}^{n-p} & \longrightarrow & \dots & \longrightarrow & 0 \\
 \downarrow & & \downarrow & & \downarrow \\
 K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r+1} \\
 \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \dots & \longrightarrow & K_{p-r}
 \end{array}$$

The last diagram is $\partial_r(x) \in E_1^{n-1, p-r}$, and the previous diagram Eq. (25) is the evidence that we can lift $\partial_r(x)$ to $\mathcal{E}_r^{n-1, p-r}$. $\square$

3 Manifolds with Corners and Framed Flow Categories

This section lays the geometric and categorical groundwork for our main constructions. We begin by employing a diagrammatic framework. The fundamental geometric objects in this setting are manifolds with corners, a standard generalization of smooth manifolds (see [16] for a relevant treatment). These concepts are then unified within the established theory of flow categories [10], a structure where the morphism spaces are themselves manifolds with corners and the composition law is governed by the geometry of their corner strata. The subsequent subsections will develop the precise definitions required for this approach.

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3.1 Diagrammatic framework

Let A be a set and $\mathcal{C}$ a category. An $\langle A \rangle$ diagram in $\mathcal{C}$ is a functor:

$$F: \{1 \rightarrow 0\}^A \rightarrow \mathcal{C}.$$

In the case where $C \in \mathcal{C}$ have a name (e.g. space, manifold, group), we call such functor an $\langle A \rangle$ space, etc.

Definition 3.1. Let A be a finite set and $a \in A$ then we define its **A -boundary** to be the restriction along the inclusion

$$i: \text{Fun}(A \setminus \{a\}, \{1 \rightarrow 0\}) \rightarrow \text{Fun}(A, \{1 \rightarrow 0\})$$

where i is given by taking the full subcategory spanned by functors with $F(a) = 1$. We get a map:

$$\partial_a: \text{Fun}(\{1 \rightarrow 0\}^A, \mathcal{C}) \rightarrow \text{Fun}(\{1 \rightarrow 0\}^{A \setminus \{a\}}, \mathcal{C})$$

An $\langle A \rangle$ -**map** is a natural transformation between $\langle A \rangle$ -diagrams.

Remark 3.2. Notice that

$$\text{Fun}(1, \text{Fun}(\{1 \rightarrow 0\}^A, \mathcal{C})) \simeq \text{Fun}(\{1 \rightarrow 0\}^A, \text{Fun}(1, \mathcal{C}))$$

hence any $\langle A \rangle$ -map is an $\langle A \rangle$ diagram in the arrow category $\text{Fun}(1, \mathcal{C})$, and we can use $\langle A \rangle$ diagrams notations, like

$$\partial_a \mu$$

Definition 3.3. Let $\mathcal{C}$ be a monoidal category. For $i \in \{1, 2\}$, let A_i be finite sets and F_i an $\langle A_i \rangle$ diagram. Then there is a tensor product

$$F_1 \otimes F_2(B_1 \sqcup B_2) := F_1(B_1) \otimes F_2(B_2).$$

which is an $\langle A_1 \sqcup A_2 \rangle$ diagram in $\mathcal{C}$. Given two morphisms $\mu_i: F_i \rightarrow G_i$ we define

$$(\mu_1 \otimes \mu_2)(B_1 \sqcup B_2) := \mu_1(B_1) \otimes \mu_2(B_2)$$

which is a map of $\langle A_1 \sqcup A_2 \rangle$ diagrams.

Taking $\mathcal{C} = \mathbf{Top}$ and let A be a finite set, here are few useful constructions that we will use later. We remind the reader that the one point compactification is a monoidal functor from locally compact Hausdorff spaces with proper maps with disjoint union (resp. Cartesian product) to pointed spaces with wedge sum (resp. smash product). Moreover it is a contravariant functor from the category of topological spaces with open inclusions, also known as the Pontrjagin-Thom collapse map. Explicitly, given $f: X \rightarrow Y$ and map continuous map between topological spaces the collapse map is the map $Y_+ \rightarrow X_+$

$$y \mapsto \begin{cases} x & y = f(x) \\ \infty_X & \text{otherwise} \end{cases}$$

Let X be $\langle A \rangle$ diagram in $\mathbf{Top}$ such that all the maps are proper. We have an $\langle A \rangle$ diagram, X_+ , given by

$$X_+(B) := (X(B))_+$$

which arising from the functoriality.

Definition 3.4. Let $f: X \rightarrow Y$ be a map of $\langle A \rangle$ diagrams in $\mathbf{Top}$ such that $f(B)$ is open inclusion, then the functoriality of $(-)_+$ yields the **collapse map**

$$f_+: Y_+ \rightarrow X_+$$

Definition 3.5. Let A be a finite set and M, N two $\langle A \rangle$ manifolds, we say that a map $e: M \rightarrow N$ is **neat embedding** if

1. For each $B \subseteq A$, the map $e(B): M(B) \rightarrow N(B)$ is an embedding.
2. The intersections $M(B) \cap N(C)$ are transverse for all $B \subseteq C$.
3. The preimage of the faces satisfies $e^{-1}(\partial_a N) = \partial_a M$ for all $a \in A$.

For $i \in \{1, 2\}$ let A_i be finite sets and let $e_i: M_i \rightarrow N_i$ be an two neat embeddings of $\langle A_i \rangle$ diagrams of manifolds. Taking the monoidal structure $(\mathbf{Top}, \times)$ we can use [Definition 3.3](#) and get a map

$$e_1 \times e_2: M_1 \times M_2 \rightarrow N_1 \times N_2$$

which is a map of $\langle A_1 \sqcup A_2 \rangle$ diagrams.

Moreover, we can define the **normal bundle**, ν_{e_i} , by

$$(\nu_{e_i})(B) := \nu_{e_i}(B)$$

which are $\langle A \rangle$ diagrams in **Top** of the total spaces of the bundles. We can do the same for the map $e_1 \times e_2$ and the fact that the total space of normal bundle commute with Cartesian product of the base, yields:

$$\nu_{e_1} \times \nu_{e_2} = \nu_{e_1 \times e_2}$$

as $\langle A_1 \sqcup A_2 \rangle$ diagrams.

Lastly, we have open inclusions of diagrams:

$$\nu_{e_i} \rightarrow N_i, \quad \nu_{e_1} \times \nu_{e_2} \rightarrow N_1 \times N_2$$

hence we can compactify and get maps

$$(N_i)_+ \rightarrow (\nu_{e_i})_+, \quad (N_1 \times N_2)_+ \rightarrow (\nu_{e_1} \times \nu_{e_2})_+$$

and the monoidality of $(-)_+$ give rise to a map of $\langle A_1 \sqcup A_2 \rangle$ diagrams:

$$(N_1)_+ \wedge (N_2)_+ \rightarrow (\nu_{e_1})_+ \wedge (\nu_{e_2})_+.$$

Definition 3.6. Given an $\langle A \rangle$ vector bundle over an $\langle A \rangle$ manifold, i.e. an $\langle A \rangle$ map $\pi : E \rightarrow M$, such that $\pi(B)$ is vector bundle for $A \subseteq A$, a **trivialization** of E is a map of $\langle A \rangle$ -spaces

$$\varphi : E \simeq M \times \mathbb{R}^N$$

for some $N \in \mathbb{N}$, where $\mathbb{R}^N$ is the $\langle \emptyset \rangle$ diagram on $\mathbb{R}^N$. This map amounts to family of bundle maps

$$\varphi(B) : E(B) \simeq M(B) \times \mathbb{R}^N.$$

We can project to the second coordinate of the RHS and get a map

$$E \rightarrow \underline{\mathbb{R}^N},$$

where the RHS is the constant functor on $\mathbb{R}^N$. Applying $(-)_+$ resulting an $\langle A \rangle$ -map

$$E_+ \rightarrow (\underline{\mathbb{R}^N})_+ \simeq \underline{S^N}. \tag{26}$$

Our main application is the case where we have a neat embedding of $\langle A \rangle$ manifolds

$$e : M \hookrightarrow \mathbb{R}^N \times \mathbb{R}_{0 \leq}^A$$

and a framing of the normal bundle

$$\varphi : \nu_e \simeq M \times \mathbb{R}^{N+|A|-n}.$$

We construct the **Thom map**:

$$f_{e,\varphi} : (\mathbb{R}^N \times \mathbb{R}_{0 \leq}^A)_+ \rightarrow (\nu_e)_+ \rightarrow (\mathbb{R}^{N+|A|-n})_+$$

where the first map is the collapse map of e and the second map is the projection of φ to the second coordinate, and applying $(-)_+$.

If $|A| = n$ we can write

$$f : S^N \wedge (\mathbb{R}_{0 \leq}^A)_+ \rightarrow (\nu_e)_+ \rightarrow S^N. \tag{27}$$

3.2 Manifolds with corners

Let $\mathbb{R}_{0\leq}$ denote the half-line:

$$\mathbb{R}_{0\leq} := [0, \infty).$$

Definition 3.7. An n -dimensional **manifold with corners** is a topological space M together with an atlas, modeled on open subsets of $\mathbb{R}_{0\leq}^n$ with smooth transition functions. We define $c(m)$ as the number of coordinates equal to zero in a chart around m . A **connected face** of M is a closures of connected components of:

$$\{m \in M \mid c(m) = 1\}.$$

A **face** is a union of connected faces.

An n -dimensional **manifolds with faces** is an n -dimensional manifold with corners such that every $m \in M$ belong to $c(m)$ connected faces of M .

Note that a face of an n -dimensional manifolds with faces is an $(n - 1)$ -dimensional manifolds with faces.

Definition 3.8. Let A be a finite set. An $\langle A \rangle$ -**manifold** is a manifold with faces M , together with a collection of faces $\{\partial_a M\}_{a \in A}$ satisfying the following conditions:

1. The union of all faces covers the boundary of M :

$$\bigcup_{a \in A} \partial_a M = \partial M.$$

2. For $a \neq b$, the intersection $\partial_a M \cap \partial_b M$ is a face of both $\partial_a M$ and $\partial_b M$.

If $A = \{1, \dots, k\}$, we refer to M as a $\langle k \rangle$ -**manifold**.

Note. Using the definitions of $\langle A \rangle$ -manifold, we get that $\partial_a M$ inherit a structure of an $\langle A \setminus a \rangle$ -manifold.

Example 3.9. The space $\mathbb{R}_{0\leq}^A$ admits an $\langle A \rangle$ -manifold structure by defining the faces:

$$\partial_a \mathbb{R}_{0\leq}^A = \mathbb{R}_{0\leq}^{A \setminus \{a\}} = \{x \in \mathbb{R}_{0\leq}^A \mid x_a = 0\}.$$

More generally, $\mathbb{R}^N \times \mathbb{R}_{0\leq}^A$ can be given an $(N + |A|)$ -dimensional $\langle A \rangle$ -manifold structure by setting:

$$\partial_a(\mathbb{R}^N \times \mathbb{R}_{0\leq}^A) = \mathbb{R}^N \times \partial_a \mathbb{R}_{0\leq}^A.$$

Example 3.10. Consider a triangle in the plane. It is a 2-dimensional manifold with corners, which admits a $\langle 3 \rangle$ -manifold structure by taking the three edges as the faces $\partial_a M$. This example illustrates that not all k -manifolds are necessarily k -dimensional. However, in this paper, we focus on manifolds where the dimension equals the number of faces, i.e., $n = k$.

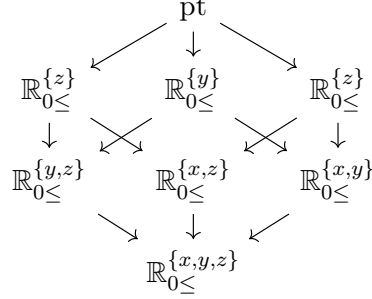
Construction 3.11. Let M be an n -dimensional $\langle A \rangle$ -manifold. It induces an $\langle A \rangle$ -diagram in the category of manifolds with embeddings by defining:

$$M(B) = \bigcap_{b \in B} \partial_b M,$$

for each subset $B \subseteq A$. In particular, $M(\emptyset) = M$, and $M(B)$ is an $(n - |B|)$ -dimensional $\langle B \rangle$ -manifold.

Remark. For simplicity, we sometimes refer to an $\langle A \rangle$ -manifold by its associated $\langle A \rangle$ -diagram.

Example 3.12. The manifold $\mathbb{R}_{0\leq}^3$ induces the following $\langle 3 \rangle$ -diagram:



The top vertex represents the origin. The second row consists of the coordinate axes $\mathbb{R}_{0\leq}^{\{x\}}$, $\mathbb{R}_{0\leq}^{\{y\}}$, and $\mathbb{R}_{0\leq}^{\{z\}}$. The third row contains the coordinate planes, which correspond to the faces $\partial_a \mathbb{R}_{0\leq}^3$. The bottom vertex is the entire manifold $\mathbb{R}_{0\leq}^3$.

Construction 3.13. For $i \in \{1, 2\}$, let M_i be an n -dimensional $\langle A \rangle_i$ -manifold, then the Cartesian product $M_1 \times M_2$ admits an $\langle A_1 \sqcup A_2 \rangle$ -manifold structure by

$$\partial_a(M_1 \times M_2) := \begin{cases} (\partial_a M_1) \times M_2 & a \in A_1 \\ M_1 \times (\partial_a M_2) & a \in A_2 \end{cases}$$

Note that this construction coincides with [Definition 3.3](#).

Definition 3.14. A neat embedding of $\langle A \rangle$ -manifolds with corners is a map of topological spaces $e : M \rightarrow N$ such that the induces map of $\langle A \rangle$ -diagrams is neat embedding.

Definition 3.15. Let M be an n dimensional $\langle A \rangle$ manifold, $e : M \hookrightarrow \mathbb{R}^N \times \mathbb{R}_{0\leq}^A$, a neat embedding. Note that the normal bundle is an $\langle A \rangle$ vector bundle over M , of rank $N + |A| - n$. A **normal framing** is a trivialization (see [Definition 3.6](#)) of ν_e

$$\varphi : \nu_e \simeq M \times \mathbb{R}^{N+|A|-n}.$$

We denote by

$$Th(M, e) := (\nu_e)_+$$

the **Thom space** of M and e , which is an $\langle A \rangle$ space.

If the choice of neat embedding is clear, we will omit e and write $Th(M)$.

Remark. Let n be the dimension of M , each normal bundle $\nu_e M(B)$ is of rank $N + |A| - n$.

Similar to [Construction 3.13](#), for $i \in \{1, 2\}$, let M_i be an embedded manifold with normal framing φ_i . Since the normal bundle of a Cartesian product is the direct sum of the individual normal bundles, the Cartesian product $M_1 \times M_2$ admits a normal framing. We denote the resulting framing by $\varphi_1 \oplus \varphi_2$.

3.3 Framed Flow Categories

We now introduce framed flow categories, first developed by Cohen, Jones, and Segal [10] to axiomatize the geometric data arising from Morse-Smale functions and certain Floer-theoretic setups. A flow category is a graded category whose morphism spaces are manifolds with corners, with dimensions determined by the grading. Composition is realized geometrically by identifying the boundary strata of one morphism manifold with the products of others.

A *framed* flow category further equips these morphism manifolds with a stable framing that must be compatible with the composition law. This coherent structure is what allows for the construction of a stable homotopy type. Our treatment, which relies on neat embeddings of these manifolds into Euclidean space, follows the perspective used in modern applications like [18].

Definition 3.16. Let $\mathcal{C}$ be a category with a **grading function**:

$$|\cdot| : \text{ob}(\mathcal{C}) \rightarrow \mathbb{Z}.$$

For each pair of objects $x, y \in \mathcal{C}$, define the linearly ordered set:

$$J_{y,x} := \{i \in \mathbb{Z} \mid |y| > i > |x|\}. \quad (28)$$

Definition 3.17. A **flow category** is a topologically enriched category $\mathcal{C}$ with a discrete set of objects and a grading function

$$|\cdot| : \text{ob}(\mathcal{C}) \rightarrow \mathbb{Z}$$

Moreover, for each pair $x, y \in \mathcal{C}$ the morphism space $\text{Hom}_{\mathcal{C}}(y, x)$ is endowed with a structure of an $|J_{y,x}|$ -dimensional $\langle J_{y,x} \rangle$ -manifold, also denoted by $M_{y,x}$, such that the corner structure satisfies:

$$\partial_i M_{z,x} = \bigsqcup_{y \in \text{ob}(\mathcal{C}), |y|=i} M_{z,y} \times M_{y,x}, \quad (29)$$

for all $i \in J_{z,x}$. The composition is given by the inclusion of the boundary:

$$M_{z,y} \times M_{y,x} \hookrightarrow \partial_i M_{z,x} \hookrightarrow M_{z,x}.$$

Notation 3.18. Let $\mathcal{C}$ be a flow category and $z, y, x \in \mathcal{C}$, we will write $\partial_y M_{z,x}$ to denote $M_{z,y} \times M_{y,x}$ as connected component of the the face $\partial_i M_{z,x}$.

Notation 3.19. Let $\mathcal{C}$ be a flow category and $j, i \in \mathbb{Z}$ we denote

$$M_{i,j} := \bigsqcup_{\substack{y \in \mathcal{C}, |y|=j \\ x \in \mathcal{C}, |x|=i}} M_{y,x}$$

Example 3.20. Consider a framed flow category $\mathcal{C}$ with object set $\{z, y, x, w\}$ and grading $|z| = 4, \dots, |w| = 1$. The morphism spaces are:

- Zero-dimensional manifolds $M_{z,y}$, $M_{y,x}$, $M_{x,w}$, which are discrete points (possibly with signs).
- One-dimensional framed manifolds $M_{z,x}$ and $M_{y,w}$, serving as null-bordisms:

$$\partial_3 M_{z,x} = M_{z,y} \times M_{y,x}, \quad \partial_2 M_{y,w} = M_{y,x} \times M_{x,w}.$$

- A two-dimensional manifold $M_{z,w}$ with faces:

$$\partial_3 M_{z,w} = M_{z,y} \times M_{y,w}, \quad \partial_2 M_{z,w} = M_{z,x} \times M_{x,w},$$

and corners given by $M_{z,y} \times M_{y,x} \times M_{x,w}$.

Definition 3.21. A neat embedding of a flow category $\mathcal{C}$ consists of a choice of integer N and a family of neat embeddings of manifolds with corners,

$$e_{y,x}: M_{y,x} \hookrightarrow \mathbb{R}^{N(|y|-|x|)} \times \mathbb{R}_{0 \leq}^{J_{y,x}},$$

for each pair of objects y, x , such that the following diagram of $\langle J_{z,x} \setminus \{|y|\} \rangle$ -spaces, commutes

$$\begin{array}{ccc} M_{z,y} \times M_{y,x} & \xrightarrow{e_{z,y} \times e_{y,x}} & \mathbb{R}^{N(|z|-|y|)} \times \mathbb{R}_{0 \leq}^{J_{z,y}} \times \mathbb{R}^{N(|y|-|x|)} \times \mathbb{R}_{0 \leq}^{J_{y,x}} \\ \downarrow & & \downarrow \\ \partial_y M_{z,x} & \xrightarrow{\partial_i e_{z,x}} & \mathbb{R}^{N(|z|-|x|)} \times \partial_i \mathbb{R}_{0 \leq}^{J_{z,x}} \end{array}$$

where $i = |y|$, the bottom map is the restriction of $\partial_i e_{z,x}$ to the y component the top map is [Definition 3.3](#) and the vertical maps are diffeomorphisms.

Note that is a uniform version, in the sense that all the manifolds embeds into a products of $\mathbb{R}^N$. This version arises by stabilizing [\[16\]](#).

Remark 3.22. The fact that any manifold with corners admits a neat embedding implies that any flow category can be embedded, see [\[16\]](#) Lemma 3.16. Moreover, the space of such embeddings becomes arbitrarily connected as N increases.

do we need this?

We recall that for two bundles $E_i \rightarrow M_i$, we have the following.

$$Th(E_1 \boxtimes E_2) \simeq Th(E_1) \wedge Th(E_2)$$

where the LHS is the Thom space of the bundle over $M_1 \times M_2$. Using the functoriality of the collapse map ([Definition 3.4](#), [Definition 3.4](#)), and the fact above, we get a commutative diagram of $\langle A \rangle$ -spaces

$$\begin{array}{ccc} S^{N(|z|-|x|)} \wedge (\mathbb{R}_{0 \leq}^{J_{z,y}} \times \mathbb{R}_{0 \leq}^{J_{y,x}})_+ & \longrightarrow & Th(M_{z,y}) \wedge Th(M_{y,x}) \xrightarrow{\simeq} Th(M_{z,y} \times M_{y,x}) \\ \downarrow & & \downarrow \\ S^{N(|z|-|x|)} \wedge (\partial_i \mathbb{R}_{0 \leq}^{J_{z,x}})_+ & \longrightarrow & \partial_y Th(M_{z,x}) \end{array} \quad (30)$$

Definition 3.23. Let $\mathcal{C}$ be an embedded flow category. A normal **framing** of $\mathcal{C}$ is a choice of normal framing

$$\varphi_{y,x}: \nu M_{y,x} \simeq M_{y,x} \times \mathbb{R}^{N(|y|-|x|)}$$

for each two objects $x, y \in \mathcal{C}$, with the following coherency

$$\partial_y \varphi_{z,x} = \varphi_{z,y} \oplus \varphi_{y,x}.$$

The compatibility of the neat embedding and the framing implies the commutativity of the diagram:

$$\begin{array}{ccccc}
 S^N(|z|-|x|) \wedge (\mathbb{R}_{0\leq}^{J_{z,y}} \times \mathbb{R}_{0\leq}^{J_{y,x}})_+ & \longrightarrow & Th(M_{z,y} \times M_{y,x}) & \xrightarrow{\varphi_{z,y} \oplus \varphi_{y,x}} & \underline{S^N(|z|-|y|)} \wedge \underline{S^N(|y|-|x|)} \\
 \downarrow & & \simeq \downarrow & & \downarrow \text{id} \\
 S^N(|z|-|x|) \wedge (\partial_i \mathbb{R}_{0\leq}^{J_{z,x}})_+ & \longrightarrow & Th(\partial_y M_{z,x}) & \xrightarrow{\partial_y \varphi_{z,x}} & \underline{S^N(|z|-|x|)}.
 \end{array} \quad (31)$$

where the vertical maps are the "Thom" maps; see Eq. (27).

3.4 Flow Modules

Definition 3.24. Let $\mathcal{C}$ be a flow category. A **right flow module** W of **degree** n over $\mathcal{C}$ consists of a collection of manifolds with corners $\{W_x\}_{x \in \mathcal{C}}$ together with compatible corner structure maps (embeddings of codimension 1 boundary faces):

$$W_y \times M_{y,x} \hookrightarrow W_x$$

for all $y, x \in \mathcal{C}$, such that:

$$\dim W_x = n - |x|.$$

Definition 3.25. Let $\mathcal{C}$ be a flow category. A **left flow module** V of **degree** p over $\mathcal{C}$ consists of a collection of manifolds with corners $\{V_x\}_{x \in \mathcal{C}}$ together with compatible corner structure maps (embeddings):

$$M_{y,x} \times V_x \hookrightarrow V_y$$

for all $y, x \in \mathcal{C}$, such that:

$$\dim V_x = |x| - p.$$

Definition 3.26. Let $\mathcal{C}$ be a neatly embedded, normally framed flow category. A **neatly embedded, normally framed right flow module** W of **degree** n over $\mathcal{C}$ consists of a right flow module structure together with a choice of integer N , neat embeddings of manifolds with corners:

$$e_x : W_x \hookrightarrow \mathbb{R}^{N(n-|x|)} \times \mathbb{R}_{0\leq}^{J_{x,-}}$$

and stable normal framings φ_x on the normal bundles νe_x that are compatible with the actions:

$$\partial_y \varphi_x = \varphi_y \oplus \varphi_{y,x}$$

where $\varphi_{y,x}$ is the framing on the morphism manifold $M_{y,x}$ of $\mathcal{C}$.

Definition 3.27. Let $\mathcal{C}$ be a framed flow category, and let $n, p, r \in \mathbb{Z}$ with $r \geq 0$. Let $W_p = \{W_z\}_{|z|=p}$ be a collection of closed framed manifolds representing an element in:

$$E_1^{n,p} = \bigoplus_{|z|=p} \Omega_{n-p}^{\text{fr}}.$$

We say that the collection W_p **extends to a right flow module of degree** n **over** $\mathcal{C}_{[p,p-r]}$ if there exists a neatly embedded, normally framed right flow module W of degree n over $\mathcal{C}_{[p,p-r]}$ whose restriction to the objects of degree p satisfies $W_z = (W_p)_z$ for all $z \in \mathcal{C}_p$.

Example 3.28. Let $\mathcal{C}$ be a framed flow category with objects $\{y_i\}$ and $\{x_j\}$ concentrated in only two degrees, $|y_i| = q$ and $|x_j| = p$. The morphism space:

$$M = \bigsqcup_{i,j} M_{y_i, x_j}$$

is a closed framed manifold of dimension $q - p - 1$, since there are no intermediate objects with degrees between p and q . In this case, $\mathcal{C}$ defines an element in the framed bordism group $\Omega_{q-p-1}^{\text{fr}}$.

4 Chain complex of Flow Category

In this section, we establish a connection between chain complexes and framed flow categories by introducing a topologically enriched category $\mathcal{J}$, which serves as a replacement for the category of chain complexes Ch . This idea originates from the work of Cohen, Jones, and Segal [10], and is further developed in the revised version [9].

The Topologically Enriched Category $\mathcal{J}$

We begin by defining the topologically enriched category $\mathcal{J}$, which is the indexing category of chain complexes in the topologically enriched setting.

Definition 4.1. We define the pointed topologically enriched category $\mathcal{J}$ as follows:

- The objects of $\mathcal{J}$ are the integers together with a distinguished object ∞ :

$$\text{ob}(\mathcal{J}) = \mathbb{Z} \cup \{\infty\}.$$

- The morphism spaces $\text{Hom}_{\mathcal{J}}(j, i)$ are defined for $i, j \in \mathbb{Z} \cup \{\infty\}$ by:

$$\text{Hom}_{\mathcal{J}}(j, i) = \begin{cases} \{\text{id}, \infty\} & \text{if } i = j, \\ \{d_i, \infty\} & \text{if } j = i + 1, \\ (\mathbb{R}_{0 \leq}^{J_{j,i}})_+ & \text{if } j > i + 1, \\ \{\infty\} & \text{if } i = \infty \text{ or } j = \infty, \end{cases}$$

where X_+ denotes the one-point compactification of a space X , and $\mathbb{R}_{0 \leq}$ is the half-line $[0, \infty)$. The set $J_{j,i}$ is defined by:

$$J_{j,i} = \{\ell \in \mathbb{Z} \mid j > \ell > i\}. \quad (32)$$

The spaces $\text{Hom}_{\mathcal{J}}(j, i)$ are pointed at the point ∞ .

- The map

$$J_{k,j} \sqcup J_{j,i} \hookrightarrow J_{k,i}$$

induce a map

$$\mathbb{R}_{0 \leq}^{J_{k,j}} \times \mathbb{R}_{0 \leq}^{J_{j,i}} \hookrightarrow \mathbb{R}_{0 \leq}^{J_{k,i}}.$$

By taking $(-)_+$ we get the composition in $\mathcal{J}$:

$$(\mathbb{R}_{0 \leq}^{J_{k,j}})_+ \wedge (\mathbb{R}_{0 \leq}^{J_{j,i}})_+ \rightarrow (\mathbb{R}_{0 \leq}^{J_{k,i}})_+.$$

this definition not worthwhile entire subsection

Remark 4.2. Intuitively, the morphisms in $\mathcal{J}$ correspond to the differentials and higher compositions in a chain complex. The space $\mathbb{R}_{0\leq}^{J_{j,i}}$ models the (higher) nullhomotopies between compositions of differentials.

We call a functor $\text{Fun}_+(\mathcal{J}, \mathbf{Top}_+)$ a **topological chain complex**.

4.1 Topological chain complexes

Our goal is to start with an embedded framed flow category and produce a topological chain complex. The idea is to use [Eq. \(27\)](#) in a compatible way to "collapse" the flow category and then to use $\mathcal{J}$ to encode maps between spheres and the higher homotopies.

Construction 4.3. Let $\mathcal{C}$ be a flow category with a neat embedding e and normal framing φ . Assume that $\mathcal{C}$ have bounded index $b \geq |-| \geq a$, without loss of generality we have $a = 0$. We define a pointed topologically enriched functor $K_{\mathcal{C}} \in \text{Fun}_+(\mathcal{J}, \mathbf{Top}_+)$ in the following way.

First

$$K_{\mathcal{C}}(i) = \bigvee_{|x|=i} S^{Nb}.$$

To define the morphisms, we use that $\text{Hom}, \wedge$ adjunction and get that we need to define maps:

$$\text{Hom}_{\mathcal{J}}(j, i) \wedge \bigvee_{|y|=j} S^{Nb} \rightarrow \bigvee_{|x|=i} S^{Nb},$$

hence it's suffice to provide maps:

$$g_{y,x} : \text{Hom}_{\mathcal{J}}(j, i) \wedge S^{Nb} \rightarrow S^{Nb}.$$

For each $y, x \in \mathcal{C}$ with indices j, i .

Let y, x be as above, we define

$$S^{N(|y|-|x|)} \wedge (\mathbb{R}_{0\leq}^{J_{y,x}})_+ \rightarrow Th(e_{y,x}) \rightarrow S^{N(|y|-|x|)}.$$

where the first map is the collapse map of $e_{y,x}$ and the second is (the compactification of) the projection of $\varphi_{y,x}$. See [Eq. \(27\)](#). By suspending this map, we get a map

$$g_{y,x} : S^{Nb} \wedge (\mathbb{R}_{0\leq}^{J_{y,x}})_+ \rightarrow Th(e_{y,x}) \rightarrow S^{Nb}.$$

It's remind to show the functoriality of K , which amounts to

$$\partial_y g_{z,x} = g_{y,x} \circ g_{z,y},$$

this equation holds due the compatibility of the neat embedding and the framing. See [Eq. \(31\)](#).

For a flow category $\mathcal{C}$, we define $\mathcal{C}^{[b,a]}$ to be the full subcategory generated by $c \in \mathcal{C}$ such that $b \geq |x| \geq a$. We denote by $K_{[b,a]}$ the topological chain complex of $\mathcal{C}^{[b,a]}$. Note that there is natural transformation

$$\beta_{b+1} : K_{[b,a]} \rightarrow K_{[b+1,a]}$$

which induces from suspending further, to $N(b+1)$, lastly if $i = b+1$ we get that $\beta_{b+1}(i)$ is the zero map $0 \rightarrow S^{N(b+1)}$.

Similarly we have maps

$$\alpha_{a-1} : K_{[b,a-1]} \rightarrow K_{[b,a]}.$$

Definition 4.4. We extend [Construction 4.3](#) to unbounded flow categories by taking (co)limit

$$K_{\mathcal{C}} = \lim_a \operatorname{colim}_b K^{[b,a]}$$

Let Sp^O be the category of orthogonal spectra. This is a topologically enriched model category for the stable homotopy category, as developed in [25]. It is equipped with a symmetric monoidal smash product $\otimes$ that gives a Quillen equivalent model to the ∞ -category of spectra. In this model, the suspension spectrum functor is strongly symmetric monoidal:

$$\Sigma^\infty : (Top_+, \wedge, S^0) \rightarrow (Sp^O, \otimes, \mathbb{S})$$

Furthermore, it preserves coproducts:

$$\Sigma^\infty : (Top_+, \vee, *) \rightarrow (Sp^O, \oplus, 0)$$

Construction 4.5. Let $\mathcal{C}$ be an embedded framed flow category bounded by b and 0 , as above. Using [Construction 4.3](#) and post composition with Σ_+^∞ we construct a functor

$$K'_{\mathcal{C}} \in \operatorname{Fun}(J, \mathbf{Sp}^O),$$

with

$$K'_{\mathcal{C}}(i) = \bigoplus_{|x|=i} \mathbb{S}^{Nb}.$$

Using post composition with $\Sigma^{-Nb} : \mathbf{Sp} \rightarrow \mathbf{Sp}$ we define the chain complex of $\mathcal{C}$ to be

$$K_{\mathcal{C}} := \Sigma^{-Nb} K'_{\mathcal{C}}$$

which have $\mathbb{S}$ in each level.

Proposition 4.6. *There is an equivalence of pointed ∞ -categories:*

$$N(\operatorname{Ch}) \simeq N_{\mathbf{Top}}(\mathcal{J}),$$

where Ch is the indexing category of chain complexes, see [Definition 2.2](#) and $N, N_{\mathbf{Top}}$ are the nerve and the topological nerve.

Proof. The morphism spaces $\operatorname{Hom}_{\mathcal{J}}(j, i)$ are contractible when $j > i + 1$ (since $(\mathbb{R}_{0 \leq}^{J_{j,i}})_+$ is contractible), and $\{\operatorname{id}, \infty\} \simeq S^0$ when $i = j$, and $\{d_i, \infty\} \simeq S^0$ when $j = i + 1$. The equivalence follows by considering both Ch and $\mathcal{J}$ as models for the same ∞ -category. $\square$

Corollary 4.7. *In light of this equivalence we can start with framed flow category and obtain a coherent chain complex in spectra*

$$\operatorname{Fun}_{\mathbf{Top}}(\mathcal{J}, \mathbf{Sp}^O) \simeq \operatorname{Fun}(N_{\mathbf{Top}}(\mathcal{J}), N_{\mathbf{Top}}(\mathbf{Sp}^O)) \simeq \operatorname{Fun}(N(\operatorname{Ch}), \mathbf{Sp}) =: \operatorname{Ch}(Sp)$$

We will, by abuse of notation, identify 1-category with it's nerves when viewing them as ∞ -categories.

We will end this section by motivating the next section.

rewrite
Construction 4.3 as a
claim about
bounded flow
cat

Let $\mathcal{C}$ be a bounded, embedded, framed flow category and let $K \in \text{Fun}(\text{Ch}, \mathbf{Sp})$ be its coherent chain complex. By [Section 2](#) we get a spectral sequence $(E_r^{n,p}, d_r)$. Note that it's immediate result that

$$E_1^{n,p} = \pi_{n-p} K_p = \pi_{n-p} \bigvee_{|x|=i} \mathbb{S}\langle x \rangle$$

where $\langle x \rangle$ denote the x component of the sum. This is the analog of the Morse chain complex. However, the description of the higher pages and the differentials is obfuscated.

The first observation is to identify stable homotopy groups with framed bordism:

$$\pi_{n-p} \mathbb{S} \simeq \Omega_{n-p}^{\text{fr}} \quad (33)$$

Let $y, x \in \mathcal{C}$ be two objects with $|y| = p, |x| = p-1$. The fact that we used the Pontrjagin-Thom collapse map to define K yields that the first differential

$$d_1: \Omega_{n-p}^{\text{fr}} \langle y \rangle \rightarrow \Omega_{(n-1)-(p-1)}^{\text{fr}} \langle x \rangle$$

is given by

$$d_1(N) = N \times M_{y,x}$$

where $M_{y,x}$ is the morphism manifold between y and x in $\mathcal{C}$ and N is a representative for it's class in the bordism group. We get that N survive to the second page if there is a $n-p+1$ dimensional manifold with boundary W , such that $\partial W = M_{y,x} \times N$ which is equivalent to the existence of a flow category with 3 objects, s, y, x and morphism manifolds $M_{y,x}, N = M_{s,y}, W = M_{s,x}$.

5 Models for Chain Complexes

At this point, given a framed flow category $\mathcal{C}$, one can combine [Construction 4.3](#) and [Section 2.2](#) to construct a spectral sequence for its coherent chain complex $K_{\mathcal{C}}$, and a sufficient condition, in terms of flow categories, for an element to survive to the r -th page. Our goal in the rest of the paper is to give a flow-categorical description for the differential and hence a necessary condition for an element to survive, in terms of flow categories.

In this section, we establish a rigorous connection between the topologically enriched category $\mathcal{J}$ (introduced in [Section 4](#) as a topologically enriched framework for chain complexes) and the ∞ -categorical framework for chain complexes detailed in [Section 2.2](#). Our primary objective is to define explicit models for the category Ch that facilitate the precise extraction of geometric data from flow categories. This extraction is useful for providing a concrete geometric interpretation of the differentials within the spectral sequence constructed in. This approach will allow us to translate abstract algebraic operations on chain complexes into intuitive geometric operations on flow manifolds.

This geometric perspective provides a concrete way to handle compositions. Specifically, we will show that two composable morphisms of chain complexes, $f: X_b \rightarrow K$ and $g: K \rightarrow Y_a$, can be understood as functors defined on higher-dimensional index categories. The composition itself is then realized as a map originating from the boundary of a multi-dimensional half-space, $\partial(\mathbb{R}_{0 \leq}^{J_{b+1,a-1}})$. Central to this construction is the identification of this boundary with a sphere:

$$S^{b-a} \simeq \partial(\mathbb{R}_{0 \leq}^{J_{b+1,a-1}})_+ = \bigcup_{b \geq p \geq a} (\mathbb{R}_{0 \leq}^{J_{b+1,p}})_+ \wedge (\mathbb{R}_{0 \leq}^{J_{p,a-1}})_+$$

add ref to the "main" Th that give a geo description for the differential

Under this equivalence, the composition map into $\text{Hom}(X_b, Y_a)$ is given explicitly by the smash product of the component maps:

$$\bigcup_{b \geq p \geq a} f_{b+1,p} \wedge g_{p,a-1} : \bigcup_{b \geq p \geq a} (\mathbb{R}_{0 \leq}^{J_{b+1,p}})_+ \wedge (\mathbb{R}_{0 \leq}^{J_{p,a-1}})_+ \rightarrow \text{Hom}(X_b, Y_a)$$

5.1 Cubical and Simplicial Models

This subsection will discuss the topologically enriched model of the cube category, with a particular focus on composition within this model. This model proves especially useful when working with chain complexes Arising from framed flow categories.

Definition 5.1. For a finite set A , we define an A -**cube** to be the category:

$$[1]^A = \{0 \rightarrow 1\}^A,$$

which is the Cartesian product of copies of the category $[1] = \{0 \rightarrow 1\}$ indexed by A .

When $A = \{1, \dots, n\}$, we write $[1]^n = [1]^{\{1, \dots, n\}}$.

In this section, we will use simplicially enriched models for ∞ -categories. Useful references include [24, Tag 00JN] and [23, Section 1.1.5]. We remind the nerve functor

$$N : \mathbf{Cat} \rightarrow \mathbf{Set}_\Delta$$

and the rectification functor

$$\text{Path} : \mathbf{Set}_\Delta \rightarrow \mathbf{Cat}_\Delta$$

which is the adjoint to the coherent nerve functor and also known as $\mathfrak{C}$.

Our first goal is to describe the simplicially enriched category $\text{Path } N[1]^A$, where $N[1]^A$ is the nerve of the category $[1]^A$.

For any two objects $O, Q \in [1]^A$, the mapping simplex $\text{Hom}_{\text{Path } N[1]^A}(O, Q)$ is constructed as follows: its vertices correspond to sequences of composable morphisms from O to Q in $[1]^A$, with edges representing compositions of morphisms, 2-simplices representing associativity conditions, and so on. Each such sequence is determined by an ordered partition of the set $Q \setminus O$, indicating the order and steps in which elements are added to O to obtain Q .

We now introduce a family of simplicial sets known as **Permutohedron**, which play significant roles in various areas of mathematics, including combinatorics and homotopy theory. For more on their flow-categorical perspective, see [14], and for their spectral sequence perspective, see [7].

Definition 5.2. For any finite set A , we define the **Permutohedron** P_A as the mapping simplex:

$$P_A := \text{Hom}_{\text{Path } N[1]^A}(\emptyset, A),$$

where $\emptyset$ denotes the initial object in $[1]^A$.

The simplicial set P_A has dimension $|A| - 1$ and can be realized geometrically as a convex polytope known as the Permutohedron.

Example 5.3. An explicit illustration of P_3 is provided in Fig. 5. The vertices correspond to the ordered partitions of the set $\{1, 2, 3\}$, and the higher-dimensional simplices represent ways of composing morphisms in $[1]^3$.

take $\{1 \rightarrow 0\}$
instead of
 $[1]$?

maybe we
want $\setminus n$ to
be from 1
and not from
0 in this con-
text

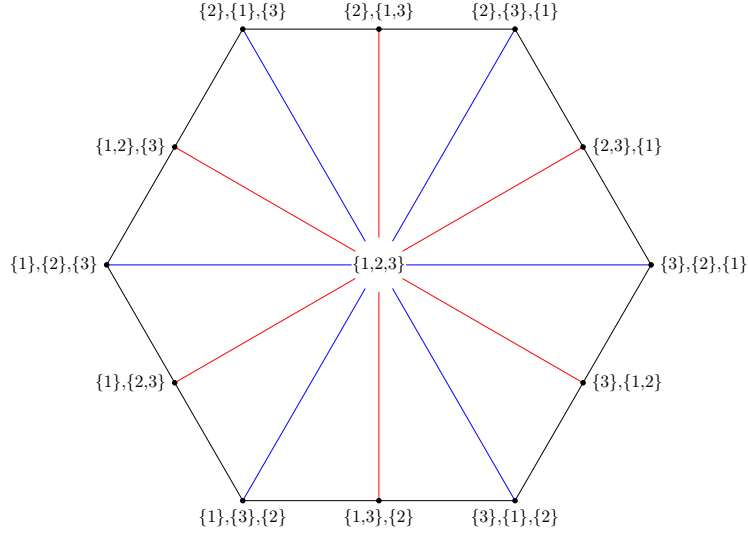


Figure 5: The mapping simplicial set $\text{Hom}_{\text{Path } N[1]^3}(\emptyset, \{1, 2, 3\})$, which is the Permutohedron P_3 . The red lines divide the hexagon into six regions representing the maximal chains from $\emptyset$ to $\{1, 2, 3\}$ in $[1]^3$. Each region has a kite shape, corresponding to the compositions of three morphisms. The blue lines subdivide the hexagon into a simplicial structure.

Lemma 5.4. *The geometric realization of the Permutohedron P_n is given by the convex hull:*

$$\|P_n\| = \text{Conv}(\{\sigma(1, 2, \dots, n)\}_{\sigma \in \Sigma_n}) \subset \mathbb{R}^n,$$

where Σ_n is the symmetric group on n elements. The simplicial structure on P_n is the barycentric subdivision of this polytope.

Proof. See [15] for details, and [14, Remark 4.9] for a flow-categorical perspective. $\square$

By considering the barycentric subdivision, each face of the polytope corresponds to a simplex in the simplicial set P_A . Vertices in the simplicial structure correspond to ordered partitions of the set A .

Remark 5.5. An important observation is that the ordered partition $(\{\sigma(1)\}, \dots, \{\sigma(n)\})$ corresponds to the point $\sigma^{-1}(1, 2, \dots, n)$ in $P_n \subset \mathbb{R}^n$. When we write $(a_1, \dots, a_n)$, we refer both to the ordered partition of size n and to the point in $\mathbb{R}^n$ corresponding to the inverse of the permutation $(a_1, \dots, a_n)$.

The polytope structure of the Permutohedron can be described in terms of ordered partitions. The faces of dimension $n - k$ correspond to ordered partitions of A into k non-empty subsets, via the inclusion of symmetric groups:

$$\Sigma_{A_1} \times \dots \times \Sigma_{A_k} \hookrightarrow \Sigma_{A_1 \sqcup \dots \sqcup A_k}.$$

In particular, the boundary of P_n can be decomposed as:

$$\partial P_n = \bigcup_{A_1 \sqcup A_2 = \{1, \dots, n\}} P_{A_1} \times P_{A_2},$$

where the union is over all ordered bipartitions of $\{1, \dots, n\}$.

encapsulate
as lemma /
prop

This decomposition reflects the composition in the category $\text{Path N}[1]^A$. For objects $O, Q, S \in [1]^A$, the composition:

$$\text{Hom}_{\text{Path N}[1]^A}(Q, S) \times \text{Hom}_{\text{Path N}[1]^A}(O, Q) \rightarrow \text{Hom}_{\text{Path N}[1]^A}(O, S)$$

corresponds to the embedding:

$$P_{S \setminus Q} \times P_{Q \setminus O} \hookrightarrow \partial P_{S \setminus O}.$$

This means that we can think of composing ordered partitions of $S \setminus Q$ and $Q \setminus O$ to obtain an ordered partition of $S \setminus O$.

5.2 Spine Cubes and Chain Complexes

To treat a morphism between chain complexes as chain complex itself we will introduce Cu_n which is a "thickening" of $\text{Ch}_{[n,0]}$.

Definition 5.6. Let A be a linearly ordered set. For each $a \in A$ we denote the subset

$$s_a = \{b \in A \mid b \leq a\} \in P(A)$$

And we call the collection $S = \{s_a\}$ the **spine** of $[1]^A = P(A)$.

Definition 5.7. Let A be a linearly ordered set, we define the **Spine cube category** Cu_A as follows

1. Start with the category $P(\mathbb{Z})^{\text{op}}$, and adjoin initial and final objects $-\infty$ and ∞ .
2. Invert the morphism $-\infty \rightarrow \infty$ to obtain a pointed category $P(A)_*^{\text{op}}$.
3. Invert all morphisms $* \rightarrow A$ for $A \notin \text{Spine}$.

The resulting category Cu_A is a quotient of $P(A)_*^{\text{op}}$.

By taking $A = \mathbb{Z}$, since all the point objects in $P(\mathbb{Z})_*^{\text{op}}$ become isomorphic in Cu , we can collapse them to a single point, obtaining an equivalent category. Therefore, we have:

Corollary 5.8. *For any $n \geq 0$, there is an equivalence of categories:*

$$\text{Cu}_{\mathbb{Z}} \simeq \text{Ch},$$

and similarly, for $b, a \in \mathbb{Z}$

$$\text{Cu}_{\{b, \dots, a\}} \simeq \text{Ch}_{[b, a]},$$

Recall that $\text{Ch}_{[b, a]}$ is the category with objects $\{b, b-1, \dots, a\}$ and a distinguished point object, as in [Definition 2.19](#).

We will write Cu for the category $\text{Cu}(\mathbb{Z})$, $\text{Cu}_{[b, a]}$ for $\text{Cu}_{\{b, \dots, a\}}$ and Cu_n for $\text{Cu}_{\{n, \dots, 0\}}$

Let $\mathcal{C}$ be a pointed ∞ -category. Combining [Corollary 5.8](#) with the n -dimensional version of [Proposition 4.6](#), we obtain three models for bounded chain complexes in $\mathcal{C}$:

$$\text{Fun}^*(\text{Cu}, \mathcal{C}) \simeq \text{Fun}^*(\text{Ch}, \mathcal{C}) \simeq \text{Fun}^*(\mathcal{J}, \mathcal{C}),$$

and similarly for the restriction to $[b, a]$.

We use the term **spine cube** when referring to a chain complex in the cubical model.

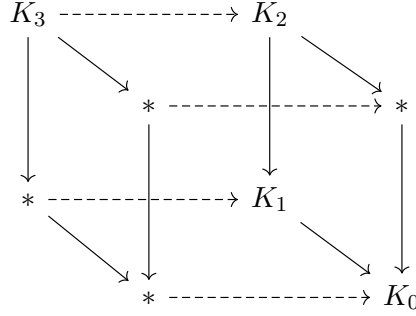


Figure 6: A spine 3-cube K , viewed as a morphism between spine 2-cubes. The dashed arrows represent morphisms between the faces (solid arrows), which are spine 2-cubes.

Definition 5.9. Let $\mathcal{C}$ be a pointed ∞ category. The inclusion $\{n\} \hookrightarrow \text{Ch}_{[n,0]}$ induces a restriction functor:

$$n_*: \text{Fun}(\text{Ch}_{[n,0]}, \mathcal{C}) \rightarrow \mathcal{C},$$

which admits a right adjoint n^* . For an object $X \in \mathcal{C}$, we define the **padded chain complex**:

$$X_n := n^*(X),$$

which is the chain complex:

$$0 \rightarrow \cdots \rightarrow 0 \rightarrow X,$$

concentrated in degree n .

Using the cubical model, we obtain:

Corollary 5.10. Let $\mathcal{C}$ be an ∞ category $X, T \in \mathcal{C}$ and $n \in \mathbb{N}$. There is an equivalence between morphisms of n -spine cubes in $\mathcal{C}$ with source of the form X_0 to $(n+1)$ -spine cubes in $\mathcal{C}$; and an equivalence between $(n+1)$ -spine cubes and morphisms with target of the form Y_n .

Proof. The equivalence

$$[1]^{n+1} \cong [1]^n \times [1]$$

induces an equivalence:

$$\text{Fun}([1]^{n+1}, \mathcal{C}) \cong \text{Fun}([1]^n \times [1], \mathcal{C}).$$

Restricting to functors that factor through $\text{Cu}_{[n,0]}$, we find that morphisms of n -chains with source X_0 correspond to $(n+1)$ -chains in $\mathcal{C}$ (see Fig. 6).

By using the equivalence $[1]^{n+1} \cong [1] \times [1]^n$ we obtain the second part of the result. $\square$

5.3 The topologically enriched model of Cu_n

Given two composable morphisms of chain complexes, we want to describe their composition in the topologically enriched model. This subsection will provide a framework to understand the composition in this model. The main output is Lemma 5.16.

Given a pointed category $\mathcal{C}$ and a functor $F \in \text{Fun}(\text{Path N Cu}_{[n,0]}, \mathcal{C})$, we observe that the boundary of the Permutohedron $P_n = \text{Hom}_{\text{Path N}[1]^n}(\emptyset, \{1, \dots, n\})$ is related to the images of the compositions:

$$P_{A_1} \times P_{A_2} \hookrightarrow \partial P_n.$$

Within this context, a specific subspace of interest is the **corner** of P_n .

Definition 5.11. We define the **corner** of P_n as the union:

$$\bigcup_{k=1}^{n-1} P_{\{0, \dots, k\}} \times P_{\{k+1, \dots, n\}} \hookrightarrow \partial P_n.$$

Notice that these are the bi-partitions such that the sets A_1, A_2 are increasing. See the bold lines in Fig. 7.

The contractibility of the spaces $\text{Hom}(*, -)$ implies that factorizations through objects not in the Spine yield trivial images. Thus, we have:

Corollary 5.12. *Let $\mathcal{C}$ be a pointed category, and let $F \in \text{Fun}(\text{Path N Cu}_n, \mathcal{C})$ be a functor. The only non-trivial part of the restriction to the boundary:*

$$\partial P_n \rightarrow \text{Hom}_{\mathcal{C}}(F(0), F(n))$$

is the corner of P_n .

Proof. Any factorization of the map $F(0) \rightarrow F(n)$ through two maps not in the Spine is trivial - due to the contractibility of the corresponding mapping spaces. Therefore, only the corner, corresponding to factorizations through the Spine, contributes non-trivially. $\square$

Definition 5.13. We define the **half cube** $\frac{1}{2}[1]^m$ as the full subcategory of $[1]^m$ consisting of subsets:

$$\{Q \subseteq \{1, 2, \dots, m\} \mid 1 \in Q \implies m \in Q\}.$$

This category is isomorphic to the iterated join $\Delta^{m-2} * [1]$.

In this category, any factorization of the map $\emptyset \rightarrow \{1, \dots, m\}$ must first include 1 and only later include m .

We also define the **diagonal** in the cube $[1]^n$ as the full subcategory generated by:

$$\{Q \subseteq \{1, 2, \dots, n\} \mid 1 \in Q \iff n \in Q\}.$$

Definition 5.14. We define the **half Permutohedron** $\frac{1}{2}P_m$ as the full subsimplicial set of P_m generated by ordered partitions in which 1 appears before m in the partition. See Fig. 7 for the case $m = 3$.

Similarly, we define the **diagonal** $\text{Diag}(P_m)$ as the $(m - 2)$ -dimensional subsimplicial set of P_m consisting of ordered partitions where 1 and m are in the same subset.

Corollary 5.15. *By restricting the factorization in $[1]^m$, we have:*

$$\text{Hom}_{\text{Path N } \frac{1}{2}[1]^m}(\emptyset, \{1, \dots, m\}) = \frac{1}{2}P_m.$$

Proof. Factorizations of the map $\emptyset \rightarrow \{1, \dots, m\}$ in $\frac{1}{2}[1]^m$ correspond to factorizations in $[1]^m$ where 1 is added before m . Therefore, the mapping simplex is the half Permutohedron $\frac{1}{2}P_m$. $\square$

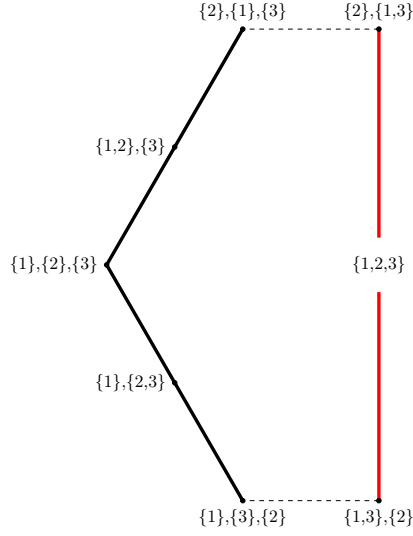


Figure 7: The half Permutohedron $\frac{1}{2}P_3$. The vertices correspond to ordered partitions where 1 appears before or together with 3. The bold lines represent the corner. The diagonal $\text{Diag}(P_3)$ is indicated in red. If we map from the category $\frac{1}{2}[1]^3$ to a category $\mathcal{C}$ such that the map is zero outside the Spine, then the dotted lines correspond to the trivial part of the boundary $\partial \frac{1}{2}P_3$.

Note that the half Permutohedron is obtained by cutting P_m along the diagonal and taking one half. Thus, we have:

$$\partial \frac{1}{2}P_m = \text{Diag}(P_m) \cup (\partial P_m \cap \frac{1}{2}P_m). \quad (34)$$

Lemma 5.16. *Let $\mathcal{C}$ be a pointed category, and let $F \in \text{Fun}(\frac{1}{2}[1]^m, \mathcal{C})$ be a functor such that $F(Q) = *$ for $Q \notin \text{Spine}$. Then there is a homotopy between the maps:*

$$\bigcup_{k=1}^{m-1} P_{\{1,\dots,k\}} \times P_{\{k+1,\dots,m\}} \rightarrow \text{Hom}_{\mathcal{C}}(F(0), F(m))$$

and

$$\text{Diag}(P_m) \rightarrow \text{Hom}_{\mathcal{C}}(F(0), F(m)),$$

as restrictions of the map:

$$\frac{1}{2}P_m \rightarrow \text{Hom}_{\mathcal{C}}(F(0), F(m)). \quad (35)$$

See Figure [Figure 7](#) for illustration.

Proof. By Equation [Equation \(34\)](#), the restriction of Equation [\(35\)](#) to the boundary of $\frac{1}{2}P_m$ decomposes as:

$$\text{Diag}(P_m) \cup (\partial P_m \cap \frac{1}{2}P_m) \rightarrow \text{Hom}_{\mathcal{C}}(F(0), F(m)).$$

Using Corollary [Corollary 5.12](#), we find that the image of the right part is non-trivial only for the corner:

$$\bigcup_{k=1}^{m-1} P_{\{1,\dots,k\}} \times P_{\{k+1,\dots,m\}}.$$

Thus, the interior of $\frac{1}{2}P_m$ provides the homotopy between these two parts. $\square$

Observation 5.17. *Note that both $\text{Diag}(P_m)$ and the corner of P_m are homeomorphic to an $(m-2)$ -dimensional disk. The lemma also implies that the boundaries of these spaces map to zero in $\text{Hom}_{\mathcal{C}}(F(0), F(m))$. Therefore, we can regard the maps:*

$$\bigcup_{k=1}^{m-1} P_{\{1, \dots, k\}} \times P_{\{k+1, \dots, m\}} \rightarrow \text{Hom}_{\mathcal{C}}(F(0), F(m))$$

and

$$\text{Diag}(P_m) \rightarrow \text{Hom}_{\mathcal{C}}(F(0), F(m))$$

as maps:

$$S^{m-2} \rightarrow \text{Hom}_{\mathcal{C}}(F(0), F(m)),$$

where S^{m-2} is the boundary of the $(m-1)$ -dimensional disk.

Remark. We remind that given bounded chain $\text{Ch}_b(\mathbf{Sp})$ we gave a description to the first page of a spectral sequence by maps $x : \mathbb{S}^{n-p} \rightarrow K_p$. Moreover such map survive to the r -th page if we can extend it to map of chains:

$$\begin{array}{ccccc} S^{n-p} & \longrightarrow & \dots & \longrightarrow & 0 \\ x \downarrow & & \downarrow & & \downarrow \\ K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r+1} \end{array}$$

Lastly, the differential $\partial_r(x)$ is given by composing:

$$\begin{array}{ccccc} S^{n-p} & \longrightarrow & \dots & \longrightarrow & 0 \\ x \downarrow & & \downarrow & & \downarrow \\ K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r+1} \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \dots & \longrightarrow & K_{p-r} \end{array} \tag{36}$$

The last

5.4 Composition in the topological model

Let $\mathcal{C}$ be a pointed ∞ -category, and let f, g be two composable maps of n -chains in $\mathcal{C}$. Using the cubical model for chains, f and g induce functors:

$$\begin{aligned} f &: \Delta^1 \times [1]^n \rightarrow \mathcal{C}, \\ g &: \Delta^1 \times [1]^n \rightarrow \mathcal{C}, \end{aligned}$$

such that their restrictions to $\partial_0 \Delta^1 \times [1]^n$ and $\partial_1 \Delta^1 \times [1]^n$ agree, allowing us to glue them along the common face:

$$\Lambda_1^2 \times [1]^n \rightarrow \mathcal{C}.$$

By the inner horn filling condition in $\mathcal{C}$, we obtain a functor:

$$\sigma \in \text{Fun}(\Delta^2 \times [1]^n, \mathcal{C}) = \text{Fun}\left(\frac{1}{2}[1]^{n+2}, \mathcal{C}\right). \tag{37}$$

combine this subsection with the prev subsec. This subsec have only one lem and also the prev subsec. Also rewrite the foreword a little bit

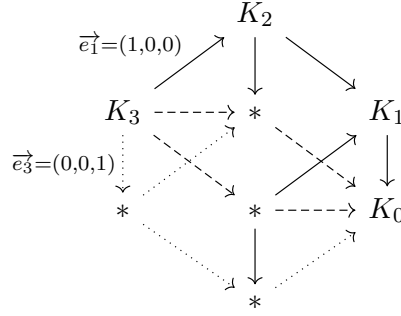


Figure 8: Composition of two n -chain maps f, g with $n = 1$, in the special case where the source of f and the target of g are padded chains X_n, Y_0 . We embed f and g in $[1]^2 \times [1]^n$. The map f corresponds to the face $x_3 = 0$, and g corresponds to $x_1 = 1$. The dashed arrows represent the diagonal (the composition), and the dotted arrows represent the rest of the cube.

For the remainder of this section, we reindex $f, g, g \circ f$, and σ according to $[1]^{n+2}$.

Assume further that f has target X_n and g has source Y_0 , where X_n, Y_0 are padded chains (see Definition [Definition 5.9](#)), which is the case when considering differentials as in Equation [Equation \(36\)](#). See Figure [Figure 8](#).

Observation 5.18. *Using the cubical model for chains, the composition:*

$$X_n \xrightarrow{f} K \xrightarrow{g} Y_0$$

is a functor:

$$[1]^{n+1} \rightarrow \mathcal{C},$$

where $\bar{0} = X$, $\bar{1} = Y$, and the rest of $[1]^{n+1}$ maps to the zero object. By taking the colimit over the diagram $[1]^{n+1} \setminus \{\bar{1}\}$, we find that $g \circ f$ is equivalent to a map:

$$\Sigma^n X \rightarrow Y$$

(see the dashed square in [Fig. 8](#) for the case $n = 1$). This map can be viewed as an element of:

$$\mathrm{Hom}_{\mathcal{C}}(X, \Sigma^{-n} Y).$$

Lemma 5.19. *Let $\mathcal{C}$ be a pointed category, and let f, g be composable maps of n -chain complexes of the form:*

$$X_n \xrightarrow{f} K \xrightarrow{g} Y_0.$$

Then the restriction of σ to the corner of P_{n+2} :

$$\bigcup_{k=1}^{n+1} P_{\{1, \dots, k\}} \times P_{\{k+1, \dots, n+2\}} \rightarrow \mathrm{Hom}_{\mathcal{C}}(X, Y)$$

is homotopic to $g \circ f$, as maps $S^n \rightarrow \mathrm{Hom}_{\mathcal{C}}(X, Y)$ (see [Observation 5.17](#)).

Proof. The composition $g \circ f$ is given by:

$$g \circ f: \partial_1 \Delta^2 \times [1]^n \rightarrow \mathcal{C}.$$

Reindexing $\Delta^2 \rightarrow [1]^2$ as a half $(n+2)$ -cube, we find that the composition corresponds to a map from:

$$\partial_1 \Delta^2 \times [1]^n = \text{Diag}([1]^{n+2}).$$

The maps f and g are functors:

$$f \in \text{Fun}([1]^{\{1, \dots, n+1\}}, \mathcal{C}), \quad g \in \text{Fun}([1]^{\{2, \dots, n+2\}}, \mathcal{C}).$$

For each $1 \leq k \leq n+1$, we obtain maps:

$$\begin{aligned} f_{0,k} &: P_{\{1, \dots, k\}} \rightarrow \text{Hom}_{\mathcal{C}}(f(0), f(k)), \\ g_{k,n+2} &: P_{\{k+1, \dots, n+2\}} \rightarrow \text{Hom}_{\mathcal{C}}(g(k), g(n+2)). \end{aligned}$$

The corner of P_{n+2} maps to $\mathcal{C}$ via:

$$\bigcup_{k=1}^{n+1} P_{\{1, \dots, k\}} \times P_{\{k+1, \dots, n+2\}} \xrightarrow{f_{0,k} \times g_{k,n+2}} \text{Hom}_{\mathcal{C}}(f(0), g(k)) \times \text{Hom}_{\mathcal{C}}(g(k), g(n+2)) \rightarrow \text{Hom}_{\mathcal{C}}(f(0), g(n+2)).$$

Under the assumptions on f and g , the filled horn σ in Equation [Eq. \(37\)](#):

$$\sigma: \frac{1}{2}[1]^{n+2} \rightarrow \mathcal{C}$$

is trivial for $Q \notin \text{Spine}$. Applying Lemma [Lemma 5.16](#), we conclude that the restriction to the corner is homotopic to $g \circ f$. $\square$

6 Differential in a flow category

By passing the definition of the differential (see [Proposition 2.29](#)) through the topological model of chain complexes, we can start with a flow category and get a flow categorical formula for the differential. Using this description we will extract an sufficient and necessary condition for the survival of an element.

Notation 6.1. For two framed flow categories $\mathcal{C}, \mathcal{D}$ with flow manifolds $M_{-, -}, N_{-, -}$ we denote by

$$\mathcal{C} = \mathcal{D}$$

the following relation: there is a bijection

$$\begin{aligned} i: \text{ob}(\mathcal{C}) &\simeq \text{ob}(\mathcal{D}) \\ M_{x,y} &= N_{i(x), i(y)} \end{aligned}$$

Definition 6.2. Let $b, a \in \mathbb{Z}$. Let $\mathcal{C}$ and $\mathcal{D}$ be framed flow categories bounded by $[b+1, a]$ and $[b, a-1]$ respectively. We say that $\mathcal{C}$ and $\mathcal{D}$ are **composable** if

$$\mathcal{C}_{[b,a]} = \mathcal{D}_{[b,a]}$$

where $\mathcal{C}_{[b,a]}$ is the full subcategory generated by $c \in \mathcal{C}$ such that $b \geq |x| \geq a$.

Similarly two chain complexes K, L (with same bounds) are composable if K, L are composable as morphisms (see [Corollary 5.10](#)).

Note that the target of K is $K_{[b,a]}$ and the source of L is $L_{[b,a]}$ we deduce that for any composable categories $\mathcal{C}, \mathcal{D}, K_{\mathcal{C}}, K_{\mathcal{D}}$ are composable chain complexes.

Our goal is to define $\mathcal{D} \circ \mathcal{C}$ to be a flow category such that

$$K_{\mathcal{D} \circ \mathcal{C}} \simeq K_{\mathcal{D}} \circ K_{\mathcal{C}}.$$

Denote by $M_{y,x}, N_{y,x}$ the morphism manifolds of $\mathcal{C}, \mathcal{D}$, the idea is to define $\mathcal{D} \circ \mathcal{C}$ to be a flow category with objects with indices $b+1, a-1$ and one morphism manifold

$$\bigcup_{b \geq p \geq a} M_{b+1,p} \times N_{p,a-1}$$

We begin by justifying the structure of this union of manifolds.

Claim 6.3. Let $\mathcal{C}, \mathcal{D}$ be two composable flow categories indexed on $b+1, a$ and $b, a-1$. Denote by $M_{y,x}$ and $N_{y,x}$ their morphism manifolds, Then

$$\bigcup_{b \geq p \geq a} M_{b+1,p} \times N_{p,a-1}$$

is a closed framed manifold with dimension $b-a+1$.

The above claim will follow from the general following "flattening" argument.

Lemma 6.4. *Let A be a finite set, $N \in \mathbb{N}$ and let*

$$D : \{1 \rightarrow 0\}^A \setminus \underline{0} \rightarrow \mathbf{Top}$$

be a diagram such that for each $a \in A$ the restriction

$$\{1 \rightarrow 0\}^{A \setminus \{a\}} \rightarrow \{1 \rightarrow 0\}^A \rightarrow \mathbf{Top}$$

is $|A|-1$ dimensional $\langle A \setminus \{a\} \rangle$ manifold. Then the topological space

$$X := \operatorname{colim}_{\{1 \rightarrow 0\}^A \setminus \underline{0}} D$$

inherits a structure of a closed $|A|-1$ manifold. Moreover, if $D(a)$ are framed, then X inherits a framing.

We refer to such diagrams as "punctured cubes".

Proof. The idea is to use the following homeomorphism:

$$\mathbb{R}_{0 \leq}^2 \simeq \mathbb{R}_{0 \leq} \times \mathbb{R}$$

which, in polar coordinates, is given by

$$(r, \theta) \mapsto (r, 2\theta).$$

Inductively we get that for any finite set A and $a \in A$ we have

$$\mathbb{R}_{0 \leq}^A \simeq H \times \mathbb{R}^{A \setminus \{a\}}$$

which induces the homeomorphism on the boundary

$$\partial \mathbb{R}_{0 \leq}^A \simeq \mathbb{R}^{A \setminus \{a\}}, \quad (38)$$

see [Definition 3.1](#).

The fact that the restrictions of D are $\langle A \setminus \{a\} \rangle$ manifolds give them the local model of

$$\mathbb{R}_{0 \leq}^{A \setminus \{a\}}$$

hence, for $B \subset A, a \in B$, locally $D(B)$ is given by

$$\mathbb{R}_{0 \leq}^{A \setminus \{a\}}(B) = \partial_a \mathbb{R}_{0 \leq}^A(B)$$

where $\mathbb{R}_{0 \leq}^A$ is the diagram of the manifolds with corners see [Construction 3.11](#) and we view $\partial_a \mathbb{R}_{0 \leq}^A$ as a sub-diagram of $\mathbb{R}_{0 \leq}^A$. Under this identification we have that

$$\partial_a \mathbb{R}_{0 \leq}^A(B) = \mathbb{R}_{0 \leq}^A(B)$$

where $D(B)$ is locally homeomorphic to $\mathbb{R}_{0 \leq}^A(B)$.

By taking colimit:

$$\operatorname{colim}_{\{1 \rightarrow 0\}^A \setminus 0} \mathbb{R}_{0 \leq}^A = \partial(\mathbb{R}_{0 \leq}^A)$$

we get that any point of X have an neighborhood homeomorphic to an open set of $\partial \mathbb{R}_{0 \leq}^A$, to get a manifold structure on X we use [Eq. \(38\)](#) and get that X is locally given by

$$\partial(\mathbb{R}_{0 \leq}^A) \simeq \mathbb{R}^{A \setminus \{a\}} \simeq \mathbb{R}^{|A|-1}.$$

Assume further that $D(a)$ are embedded and normally framed manifolds with framing φ_a . The natural transformations φ_a

$$\nu D(a) \simeq D(a) \times \mathbb{R}^N$$

(where $N \in \mathbb{N}$) assemble into a neutral transformation

$$\nu D \simeq D \times \mathbb{R}^N$$

where the left is the diagram $\nu D(B)$ and the right is $D(B) \times \mathbb{R}^N$. By taking colimit we get a normal framing

$$\nu X \simeq X \times \mathbb{R}^N.$$

□

Example 6.5. Take $b = 2, a = 1$ and take $\mathcal{C}, \mathcal{D}$ to be flow categories with tree objects. The data of $\mathcal{C}$ amounts to two closed 0-dimensional manifolds $M_{3,2}, M_{2,1}$ and a 1-dimensional manifold $M_{3,1}$ with boundary

$$\partial M_{3,1} = M_{3,2} \times M_{2,1}$$

and similar for $\mathcal{D}$ with manifolds $N_{2,1}, N_{1,0}$ and $N_{2,0}$, where $M_{2,1} = N_{2,1}$. The construction is given by gluing

$$M_{3,1} \times N_{1,0} \cup M_{3,2} \times N_{2,0}$$

along $M_{3,2} \times M_{2,1} \times N_{1,0}$.

another approach is to factor the structure maps $D(B) \rightarrow BO(?)$ through $*$, not sure if this is too homotopical

This example recovers the secondary structure on bordism rings, introduced by Alexander in [3]. We can think of $\mathcal{D} \circ \mathcal{C}$ as the *Toda brackets* or *Massey product* of the 0-dimensional manifolds of $\mathcal{C}, \mathcal{D}$.

Definition 6.6. Let $\mathcal{C}, \mathcal{D}$ be two composable flow categories with flow manifolds $M_{-, -}, N_{-, -}$. We define

$$D: \{1 \rightarrow 0\}^{\{b, \dots, a\}} \setminus \underline{0} \rightarrow \mathbf{Top}$$

in the following way: for each $B = \{i_k, \dots, i_1\} \subseteq \{b, \dots, a\}$

$$D(i_k, \dots, i_1) = M_{b+1, i_k} M_{i_k, i_{k-1}} \times \dots \times M_{i_2, i_1} \times N_{i_1, a-1}.$$

We write

$$M_{b+1, a-1} := \bigcup_{b \leq p \leq a} M_{b+1, p} \times N_{p, a-1} := \text{colim } D.$$

The **composition** flow category $\mathcal{C} \circ \mathcal{D}$, is given by

$$\text{ob}(\mathcal{D} \circ \mathcal{C}) := \{y, x \mid y \in \mathcal{C}_{b+1}, x \in \mathcal{D}_{a-1}\}, \quad |y|_{\mathcal{D} \circ \mathcal{C}} = |y|_{\mathcal{C}} - 1, |x|_{\mathcal{D} \circ \mathcal{C}} = |x|_{\mathcal{D}}$$

and $M_{b+1, a-1}$ is the morphism manifold of this category.

Remark 6.7. In our convention $M_{b+1, a-1}$ is the union of all the morphism manifolds of $\mathcal{C} \circ \mathcal{D}$. To get the morphism manifolds $M_{y, x}$ between two objects $y, x \in \mathcal{C} \circ \mathcal{D}$, one need to modify the colimit by replacing M_{b+1, b_k} with M_{y, b_k} and $M_{b_1, a-1}$ with $M_{b_1, x}$.

Claim 6.8. Let D be a punctured diagram as in Lemma 6.4, then we have that

$$(\text{colim } D(B))_+ \simeq \text{colim}(D(B)_+)$$

Proof. Note that

$$\text{colim } D(B)_+ = \text{colim}(D(B) \cup \{\infty_B\}) \simeq (\text{colim } D(B)) \cup \{\infty_A\}$$

where that last equivalence is the identification of all the points with the point of $D(A)$. This resulting a compact space, which is differ from $\text{colim } D$ by one point, hence we get that

$$\text{colim } D(B)_+ \simeq (\text{colim } D)_+$$

□

Lemma 6.9. Let $\mathcal{C}, \mathcal{D}$ be two composable framed flow categories, then We have that

$$K_{\mathcal{D} \circ \mathcal{C}} \simeq K_{\mathcal{D}} \circ K_{\mathcal{C}}$$

Proof. First we remind that $K_{\mathcal{C}}$ is the chain complexes with maps

$$f_{z, y} : \mathbb{R}_{0 \leq}^{J_{z, y}} \wedge S^{N(b+1-a)} \rightarrow S^{N(b+1-a)}$$

which constructed by collapsing $M_{z, y}$. Similarly the chain complex $K_{\mathcal{D}}$ have the maps $g_{y, x}$ which constructed by collapsing $N_{y, x}$.

Using the topologically enriched model, by [Lemma 5.19](#) our goal is to show that

$$c\left(\bigcup_{b \geq p \geq a} M_{b+1,p} \times N_{p,a-1}\right) = \bigcup_{b \geq p \geq a} f_{b+1,p} \wedge g_{p,a-1}$$

where c is the Pontrjagin-Thom collapse map.

Let D be a punctured cube diagram as above. Let $X := \operatorname{colim} D$ be the colimit manifold. We denote

$$A = J_{b+1,a-1} = \{b, \dots, a\}$$

and note that $|A| = b - a + 1$. We remind that the Pontryagin Thom map of X is constructed in the following way: for each $B \subseteq \{b, \dots, a\}$ we have a neat embedding

$$e(B): D(B) \hookrightarrow \mathbb{R}_{0 \leq}^A(B) \times \mathbb{R}^{N(b-a+2)}$$

by taking colimit we get the assembly map

$$X = \operatorname{colim}_{\{1 \rightarrow 0\}^A \setminus \emptyset} D \hookrightarrow \operatorname{colim}_{\{1 \rightarrow 0\}^A \setminus \emptyset} \mathbb{R}_{0 \leq}^A \times \mathbb{R}^{N(b-a+2)} \simeq \mathbb{R}^{|A|-1} \times \mathbb{R}^{N(b-a+2)}.$$

We get an open embedding of the normal

$$e: \nu X \hookrightarrow \mathbb{R}^{|A|-1} \times \mathbb{R}^{N(b-a+2)}$$

which is the colimit of the maps

$$e(B): \nu D(B) \hookrightarrow \mathbb{R}_{0 \leq}^A(B) \times \mathbb{R}^{N(b-a+2)}.$$

On the other hand we have the normal framing, composed with the projection

$$\varphi(B): \nu D(B) \simeq D(B) \times \mathbb{R}^{N(b-a+2)} \rightarrow \mathbb{R}^{N(b-a+2)}$$

which assemble into the map

$$\operatorname{colim} \varphi: \nu X \rightarrow \mathbb{R}^{N(b-a+2)}$$

The Thom map (see [Eq. \(27\)](#)) of X is given by

$$c_X = \varphi_+ \circ e_+$$

where e_+ is given by collapsing the complement of νX . We can re-write this map as

$$(\operatorname{colim} \varphi(B))_+ \circ \operatorname{colim}(e(B))_+$$

and using [Claim 6.8](#) we get that

$$(\operatorname{colim} \varphi(B)_+) \circ \operatorname{colim}(e(B)_+) = \operatorname{colim}(\varphi(B)_+ \circ e(B)_+)$$

which is the colimit over the diagram of taking point-wise Thom maps of $D(B)$. To finish the proof we need to show that the Thom maps of

$$D(\{p\}) = M_{b+1,p} \times N_{p,a-1}$$

are homotopic to $f_{b+1,p} \wedge g_{p,a-1}$. This follows from the functoriality of $(-)_+$ and by isotoping the tubular neighborhoods

$$\nu(M_{b+1,p} \times N_{p,a-1}) \text{ and } \nu M_{b+1,p} \times \nu N_{p,a-1}.$$

□

Theorem 6.10. *Let $\mathcal{C}$ be a flow category with a neat embedding and normal framing. Let $K_{\mathcal{C}}$ be the chain complex obtained by collapsing the morphism manifolds of $\mathcal{C}$, and denote by $E_r^{n,p}$ the spectral sequence of $K_{\mathcal{C}}$. We have that:*

$$E_1^{n,p} = \bigoplus_{|x|=p} \Omega_{n-p}^{\text{fr}}.$$

A collection of closed framed manifolds $L = \{L_z\}_{|z|=p}$ representing an element in $E_1^{n,p}$ survives to the $(r+1)$ -th page if and only if L extends to a right flow module W of degree n over $\mathcal{C}_{[p,p-r]}$ (see [Definition 3.27](#)).

If L survives to the $(r+1)$ -th page, then the differential $d_{r+1}(L)$ is given by the bordism class of the Toda manifold $W \circ \mathcal{M}_{-,p-r-1}$ (the geometric tensor product of the right module W and the left module $\mathcal{M}_{-,p-r-1}$), which is represented by the closed framed manifold:

$$\bigcup_{p \geq q \geq p-r} W_q \times M_{q,p-r-1}$$

where the W_q are the manifolds in the flow module W , and the $M_{q,p-r-1}$ are the morphism manifolds of $\mathcal{C}$.

Proof. On the one hand, assume that the collection L extends to a right flow module W over $\mathcal{C}_{[p,p-r]}$ of degree n . Using [Construction 4.3](#), we construct a map of coherent chain complexes (or a $\mathcal{J}$ -module map) corresponding to W :

$$K_W : K_{S_n} \rightarrow K_{\mathcal{C}_{[p,p-r]}}$$

which extends the map representing L on the degree p objects. By the homological criterion established in [Proposition 2.24](#), this map of chain complexes guarantees that the element L survives to the $(r+1)$ -th page of the spectral sequence.

For the differential, let W be the right flow module of degree n over $\mathcal{C}_{[p,p-r]}$ extending L . By [Proposition 2.29](#), the differential $d_{r+1}(L)$ is given by the composition of the chain maps:

$$K_{\mathcal{M}_{-,p-r-1}} \circ K_W : K_{S_n} \rightarrow K_{p-r-1}$$

where the first map is the left flow module $\mathcal{M}_{-,p-r-1}$ viewed as a chain map. Under the equivalence between compositions of chain complexes and compositions of modules, this corresponds to the Toda manifold $W \circ \mathcal{M}_{-,p-r-1}$, which is represented by the closed framed manifold:

$$\bigcup_{p \geq q \geq p-r} W_q \times M_{q,p-r-1}$$

where the W_q are the manifolds of the module W and $M_{q,p-r-1}$ are the morphism manifolds of $\mathcal{C}$.

The remaining part is to show that if $L \in E_1^{n,p}$ survives to the $(r+2)$ -th page, then the collection L extends to a right flow module over $\mathcal{C}_{[p,p-r-1]}$. By the definition of survival, if L survives then $d_{r+1}(L) = 0$ in bordism. Thus, the Toda manifold $W \circ \mathcal{M}_{-,p-r-1}$ is null-bordant, meaning there exists a framed manifold with corners W_{p-r-1} of dimension $n - p + r$ such that:

$$\partial W_{p-r-1} = W \circ \mathcal{M}_{-,p-r-1} = \bigcup_{p \geq q \geq p-r} W_q \times M_{q,p-r-1}.$$

By adding the manifold W_{p-r-1} to the collection $\{W_q\}_{q \in \mathcal{C}_{[p,p-r]}}$, we obtain a right flow module over $\mathcal{C}_{[p,p-r-1]}$ of degree n , as required. $\square$

7 Tangential structures and change of rings

In this section we point out how to generalize [Theorem 6.10](#) to a case where we replace the framing structure on the flow category $\mathcal{C}$ by other tangential structures. As a result we get a spectral sequence that calculates the bordism homology $\Omega_*^{\mathcal{X}}(|\mathcal{C}|)$.

To start with, we will generalize the definition of normal framing.

Definition 7.1. A tangential structure is a collection of fibrations $\mathcal{X}(n) \rightarrow BO(n)$, with maps $\mathcal{X}(n) \rightarrow \mathcal{X}(n+1)$ such that

$$\begin{array}{ccc} \mathcal{X}(n) & \longrightarrow & \mathcal{X}(n+1) \\ \downarrow & & \downarrow \\ BO(n) & \longrightarrow & BO(n+1) \end{array}$$

Definition 7.2. Let $M \hookrightarrow \mathbb{R}_{0 \leq}^k \times \mathbb{R}^N$ be an embedded m dimensional $\langle k \rangle$ manifold. A **normal $\mathcal{X}$ structure on M** is a choice of a lift

$$\begin{array}{ccc} & & \mathcal{X}(N) \\ & \nearrow \nu \tilde{M} & \downarrow \\ M & \xrightarrow{\nu M} & \underline{BO(N+k-m)} \end{array} \quad (39)$$

where the diagram takes place in the category $\langle k \rangle$ spaces, which is a functorial $\{1 \rightarrow 0\}^k$ family of lifts

$$\begin{array}{ccc} & & \mathcal{X}(N) \\ & \nearrow & \downarrow \\ M(A) & \xrightarrow{\nu M(A)} & \underline{BO(N+k-m)} \end{array} \quad (40)$$

for each $A \subseteq \{1 \rightarrow 0\}^k$.

Definition 7.3. We say that an $\mathcal{X}$ structure is **multiplicative** if it is equipped with structure maps

$$\mathcal{X}(n) \times \mathcal{X}(m) \rightarrow \mathcal{X}(n+m)$$

that commutes with the maps

$$BO(n) \times BO(m) \rightarrow BO(n+m)$$

Note that if $\mathcal{X}$ is multiplicative then a choice of $\mathcal{X}$ structure on two embedded manifolds M, L induces a $\mathcal{X}$ structure on $M \times L$ in the following way

$$\begin{array}{ccccc} & & \mathcal{X}(d) \times \mathcal{X}(d') & \longrightarrow & \mathcal{X}(d+d') \\ & \nearrow & \downarrow & & \downarrow \\ L \times M & \longrightarrow & \underline{BO(d) \times BO(d')} & \longrightarrow & \underline{BO(d+d')} \end{array}$$

where d and d' are the codimensions of L and M , respectively.

For an tangential structure $\mathcal{X}$ we denote by ζ_n the universal real vector bundle over $X(n)$, which is (by abuse of notation) the pullback of the universal real vector bundle $\zeta_n \rightarrow BO(n)$. Let M be an $\mathcal{X}$ structure manifold (for simplicity - a closed manifold) and denote by d the codimension of M in the ambient space, we obtain the following diagram:

$$\begin{array}{ccccc} \nu M & \longrightarrow & \zeta_n & \longrightarrow & \zeta_n \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ M & \xrightarrow{\nu \tilde{M}} & X(d) & \longrightarrow & BO(d) \end{array}$$

where the right square is pullback by definition and the same for the whole rectangle. We denote

$$\varphi_M: \nu M \rightarrow \zeta_n \tag{41}$$

Note that although we omitted $\nu \tilde{M}$ from the notation, a different choices of $\nu \tilde{M}$ defines different map φ_M . Note that if M is $\langle k \rangle$ manifold then φ_M is a map of $\langle k \rangle$ vector bundles.

Given a multiplicative structure we can define an $\mathcal{X}$ structured flow category (compare to [Definition 3.23](#)).

Definition 7.4. Let $\mathcal{C}$ be an embedded flow category, an $\mathcal{X}$ structure on $\mathcal{C}$ is a choice of $\mathcal{X}$ structure on each embedded morphism manifold $M_{y,x}$ such that

$$\partial_y \varphi_{z,x} = \varphi_{z,y} \oplus \varphi_{y,x}$$

where $\varphi_{y,x}$ is the isomorphism of $\langle J_{y,x} \rangle$ bundles arise from the $\mathcal{X}$ structure on $M_{y,x}$, see [Eq. \(41\)](#).

Similarly to [Construction 4.3](#), using Pontryagin Thom we get a coherent chain complex in $\text{Mod}_{M\mathcal{X}}$ where $M\mathcal{X}$ is the Thom spectrum of $\mathcal{X}$.

Construction 7.5. Let $\mathcal{C}$ be a bounded $\mathcal{X}$ flow category. For each $x, y \in \mathcal{C}$ we define a map

$$g_{y,x} : S^{N(|y|-|x|)} \wedge (\mathbb{R}_{0 \leq}^{J_{y,x}})_+ \rightarrow Th(\nu M_{y,x}) \rightarrow Th(\zeta_{N(|y|-|x|)}).$$

where the first map is the collapse map of the embedding $e_{y,x}$ (see [Definition 3.4](#)) and the second map is $Th(\varphi_{y,x})$. The coherence of being $\mathcal{X}$ flow categories amounts to

$$g_{z,y} \wedge g_{y,x} = \partial_y g_{z,x}$$

as maps between diagrams in **Top**.

By taking $\Sigma^{N(|y|-|x|)} \Sigma_+^\infty g_{y,x}$ we get a collection of maps to the Thom spectrum of $\mathcal{X}$

$$\mathbb{S} \otimes (\mathbb{R}_{0 \leq}^{J_{y,x}})_+ \rightarrow M\mathcal{X}.$$

which is equivalent to a map

$$(\mathbb{R}_{0 \leq}^{J_{y,x}})_+ \rightarrow \text{Hom}_{\mathbf{Sp}}(\mathbb{S}, M\mathcal{X}),$$

the last step is to use the adjunction between $\mathbb{S} - \text{Mod}$ and $\mathcal{X} - \text{Mod}$ to obtain a map

$$\bar{g}_{y,x} : (\mathbb{R}_{0 \leq}^{J_{y,x}})_+ \rightarrow \text{Hom}_{M\mathcal{X}}(M\mathcal{X}, M\mathcal{X}).$$

To get that the collection of these maps define a functor

$$\mathcal{J} \rightarrow \text{Mod} - \mathcal{X}$$

we use the following commutative diagram:

$$\begin{array}{ccc} \text{Hom}_{\mathbf{Top}}(S^d, \zeta_d) \times \text{Hom}_{\mathbf{Top}}(S^{d'}, \zeta_{d'}) & \xrightarrow{\wedge} & \text{Hom}_{\mathbf{Top}}(S^{d+d'}, \zeta_{d+d'}) \\ (\Sigma^{-d}\Sigma_+^\infty, \Sigma^{-d'}\Sigma_+^\infty) \downarrow & & \downarrow \Sigma^{-(d+d')}\Sigma_+^\infty \\ \text{Hom}_{\mathbf{Sp}}(\mathbb{S}, M\mathcal{X}) \times \text{Hom}_{\mathbf{Sp}}(\mathbb{S}, M\mathcal{X}) & & \text{Hom}_{\mathbf{Sp}}(\mathbb{S}, M\mathcal{X}) \\ \text{Mod} \downarrow & & \downarrow \text{Mod} \\ \text{Hom}_{M\mathcal{X}}(M\mathcal{X}, M\mathcal{X}) \times \text{Hom}_{M\mathcal{X}}(M\mathcal{X}, M\mathcal{X}) & \xrightarrow{\circ} & \text{Hom}_{M\mathcal{X}}(M\mathcal{X}, M\mathcal{X}) \end{array}$$

where Mod denote the $\mathbb{S} - \text{Mod}, \mathcal{X} - \text{Mod}$ adjunction. We obtain that

$$\bar{g}_{z,y} \otimes \bar{g}_{y,x} = \partial_y \bar{g}_{z,x}$$

hence we got a (topologically enriched) coherent chain complex $\text{Fun}(\mathcal{J}, \text{Mod} - \mathcal{X})$.

The definition of a right flow module ([Definition 3.24](#)) is modified by replacing framed flow modules by $\mathcal{X}$ structured flow modules and Ω^{fr} by $\Omega^{\mathcal{X}}$.

Similarly, we define when two $\mathcal{X}$ flow categories are composable. To define the composition category we modify [Definition 6.6](#) we get a boundary diagram

$$D: \{1 \rightarrow 0\}^{\{b, \dots, a\}} \setminus \underline{0} \rightarrow \mathbf{Top}$$

such that $D(\{a\})$ are $\mathcal{X}$ manifolds with $\langle A \setminus \{a\} \rangle$ corner structure. he lifts of the structure map $\nu D(B)$ assemble into a lift of the structure map $\nu \text{colim } D$ hence induces a $\mathcal{X}$ structure on the colimit. We get that

Corollary 7.6. *Let $\mathcal{C}, \mathcal{D}$ be two composable flow categories bounded by $b+1, a-1$, with $\mathcal{X}$ structure and flow manifolds written as $M_{-, -}, N_{-, -}$ then*

$$\bigcup_{b \leq p \leq a} M_{b+1, p} \times N_{p, a-1} := \text{colim } D$$

is an $\mathcal{X}$ manifold (compare to [Definition 6.6](#)). Where D is defined in the following way: for each $B = \{i_k, \dots, i_1\} \subseteq \{b, \dots, a\}$

$$D(i_k, \dots, i_1) = M_{b+1, i_k} M_{i_k, i_{k-1}} \times \dots M_{i_2, i_1} \times N_{i_1, a-1}.$$

We write

$$M_{b+1, a-1} := \bigcup_{b \leq p \leq a} M_{b+1, p} \times N_{p, a-1} := \text{colim } D.$$

We define composition category as same as the framed case.

Theorem A. *Let $\mathcal{C}$ be a flow category with an $\mathcal{X}$ -tangential structure. There is a spectral sequence:*

$$E_1^{n,p} = \bigoplus_{|z|=p} \Omega_{n-p}^{\mathcal{X}}$$

which converges to the homotopy groups of the realization:

$$E_r^{n,p} \implies \pi_n |\mathcal{C}|.$$

An element

$$W_p = \{W_z\}_{|z|=p} \in E_1^{n,p}$$

survives to $E_{r+1}^{n,p}$ if and only if the collection W_p extends to a right flow module over $\mathcal{C}_{[p,p-r]}$ (see [Definition 3.27](#)). The differential is represented geometrically by the Toda manifolds:

$$d_{r+1}(W_p) = \bigcup_{y \in \mathcal{C}_{p-(r+1)}} (\mathcal{M}_{-,y} \circ W) \in \bigoplus_{y \in \mathcal{C}_{p-(r+1)}} \Omega_{n-p+r}^{\mathcal{X}} = E_{r+1}^{n-1,p-(r+1)}$$

7.1 Change of rings

Given a framed flow category $\mathcal{C}$, we can benefit from the fact that the changing the tangential structure of manifolds correspond to maps between spectra, to get a spectral sequence that calculates the bordism homology (represented by Thom spectra) of a manifold.

Let $\mathcal{X}, \mathcal{Y}$ be two tangential structures. Denote by $\mathbf{Flow}^{\mathcal{X}}$ (resp. $\mathbf{Flow}^{\mathcal{Y}}$) the collections of $\mathcal{X}$ (resp. $\mathcal{Y}$) structured flow categories.

A map $\mathcal{X} \rightarrow \mathcal{Y}$ induce a "forgetting" map F , from $\mathcal{X}$ manifolds to $\mathcal{Y}$ manifolds. The fact that Cartesian product commutes with F , induces a map:

$$(-)^{\mathcal{Y}} : \mathbf{Flow}^{\mathcal{X}} \rightarrow \mathbf{Flow}^{\mathcal{Y}}$$

On the other hand, the map $\mathcal{X} \rightarrow \mathcal{Y}$ induces a map of spectra $M\mathcal{X} \rightarrow M\mathcal{Y}$ and hence a map of coherent chain complexes:

$$\otimes_{M\mathcal{X}} M\mathcal{Y} : \text{Ch}(\text{Mod}_{M\mathcal{X}}) \rightarrow \text{Ch}(\text{Mod}_{M\mathcal{Y}})$$

induces by post composition

$$\text{Ch} \xrightarrow{K} \text{Mod}_{M\mathcal{X}} \xrightarrow{\otimes_{M\mathcal{X}} M\mathcal{Y}} \text{Mod}_{M\mathcal{Y}}$$

Lemma 7.7. *The following diagram commutes:*

$$\begin{array}{ccc} \mathbf{Flow}^{\mathcal{X}} & \xrightarrow{K_{(-)}} & \text{Ch}(\text{Mod}_{M\mathcal{X}}) \\ (-)^{\mathcal{Y}} \downarrow & & \downarrow \otimes_{M\mathcal{X}} M\mathcal{Y} \\ \mathbf{Flow}^{\mathcal{Y}} & \xrightarrow{K_{(-)}} & \text{Ch}(\text{Mod}_{M\mathcal{Y}}) \end{array}$$

Re-write the proof

Proof. Let $\mathcal{C}$ be a $\mathcal{X}$ flow category. On the one hand we have $K_{\mathcal{C}^{\mathcal{Y}}}$, on the other $K_{\mathcal{C}} \otimes_{M\mathcal{X}} M\mathcal{Y}$. The object of the coherent chains are:

$$(K_{\mathcal{C}^{\mathcal{Y}}})_p = \bigoplus_{|z|=p} M\mathcal{Y} = \bigoplus_{|z|=p} M\mathcal{X} \otimes_{M\mathcal{X}} M\mathcal{Y} \simeq (K_{\mathcal{C}} \otimes_{M\mathcal{X}} M\mathcal{Y})_p.$$

We have that the maps of $K_{\mathcal{C}^{\mathcal{Y}}}$ are the collapse maps:

$$c_{M_{z,w}^{\mathcal{Y}}} : \Sigma_+^{\infty} \mathbb{R}_{0 \leq}^{J_{z,w}} \rightarrow M\mathcal{Y} \quad (42)$$

where $M^{\mathcal{Y}}$ is the manifold M with $\mathcal{Y}$ structure. The maps of $K_{\mathcal{C}} \otimes_{M\mathcal{X}} M\mathcal{Y}$ are given by collapsing and tensoring

$$c_{M_{z,w}} \otimes_{M\mathcal{X}} M\mathcal{Y} : \Sigma_+^{\infty} \mathbb{R}_{0 \leq}^{J_{z,w}} \rightarrow M\mathcal{X} \xrightarrow{(-) \otimes_{M\mathcal{X}} M\mathcal{Y}} M\mathcal{Y}. \quad (43)$$

The fact that the $\mathcal{Y}$ structure of $M_{z,w}$ is coming from $\mathcal{X}$ yields that the collapse map of $M_{z,w}^{\mathcal{Y}}$ factors through $\mathcal{X}$, the functoriality of the collapse map give the equivalence between Eq. (42) and Eq. (43). $\square$

First, we get that we can start with framed flow category and forget to other tangential structure, to get that:

Corollary 7.8. *Let $\mathcal{C}$ be a framed flow category and let $\mathcal{X}$ be a tangential structure. Then the spectral sequence of Theorem A implies to $\mathcal{C}^{\mathcal{X}}$ having:*

$$E_r \implies \pi_*(|\mathcal{C}^{\mathcal{X}}|) \simeq \pi_*(|\mathcal{C}| \otimes M\mathcal{X}) =: M\mathcal{X}_*(|\mathcal{C}|)$$

The first equivalence holds because the realization from Ch to Fil is given by colimit, hence commutes with tensoring (as left adjoint).

Second, we can apply it to Morse function, to calculate the bordism homology of a manifold:

Corollary 7.9. *Let $f : M \rightarrow \mathbb{R}$ be a Smale Morse function and let $\mathcal{X}$ be a tangential structure, then the spectral sequence of the framed flow category $(\mathcal{C}_f)^{\mathcal{X}}$ converge to $M\mathcal{X}_*(M)$.*

7.2 Bordism homology of projective spaces

Let $A = MO$, the unoriented bordism spectrum. We remind that

$$MO_* = \mathbb{Z}/2 \left[x_i; i \neq 2^k - 1 \right]$$

Let $a \in \mathbb{R}^{n+1}$ be a weights vector such that $a_i < a_{i+1}$ and let $\bar{f} : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ be the weighted distance function:

$$\bar{f}(x) = \sum a_i x_i^2$$

Let $M = \mathbb{RP}^{\ell}$, notice that $\bar{f}(x) = \bar{f}(\lambda x)$, therefore $\bar{f}$ induce a function $f : M \rightarrow \mathbb{R}$.

Notice that the critical points of f are given by:

$$e_i := [0 : \cdots : 0 : \underbrace{1}_i : 0 : \cdots : 0]$$

with index:

$$|e_i| = i - 1.$$

Denote by $E_r^{n,p}$ the spectral sequence of [Corollary 7.9](#) with coefficients in MO . We get that E_1 is given by:

$$E_1^{n,p} = \Omega_{n-p}^O$$

To calculate the flow manifolds, first notice that the stable manifolds are given by:

$$W_p^s = [0 : \cdots : 0 : \underset{\neq 0}{x_p} : x_{p+1} : \cdots : x_\ell] \quad (44)$$

and the unstable manifolds are given by:

$$W_p^u = [x_0 : \cdots : x_{p-1} : \underset{\neq 0}{x_p} : 0 : \cdots : 0] \quad (45)$$

The intersection is:

$$W_q^u \cap W_p^s = [0 : \cdots : 0 : \underset{\neq 0}{x_p} : x_{p+1} : \cdots : x_{q-1} : \underset{\neq 0}{x_q} : 0 : \cdots : 0] \quad (46)$$

By solving the flow equation we get that the action of the flow is given by:

$$\phi_t([x_i]) = [e^i t x_i]$$

So we get that the flow acts on $W_q^u \cap W_p^s$ by:

$$\begin{aligned} [0 : \cdots : 0 : \underset{\neq 0}{x_p} : x_{p+1} : \cdots : \underset{\neq 0}{x_{q-1}} : x_q : 0 : \cdots : 0] &\mapsto \\ [0 : \cdots : 0 : \underset{\neq 0}{x_p} : e^t x_{p+1} : \cdots : e^{(q-p-1)t} x_{q-1} : \underset{\neq 0}{e^{(q-p)t} x_q} : 0 : \cdots : 0] \end{aligned}$$

By using the scalar multiplication in $\mathbb{RP}^\ell$ we can assume that $x_p = 1$. Moreover, by taking quotient by the flow ϕ_t we can assume that in the quotient:

$$W_q^u \cap W_p^s / \phi_t$$

the q coordinate is ± 1 . We conclude that the (non compactified) flow manifolds are:

$$M'_{q,p} = \{[0 : \cdots : 0 : 1 : x_{p+1} : \cdots : x_{q-1} : \pm 1 : 0 : \cdots : 0]\}$$

And we get that after Compactification

Proposition 7.10. *The flow manifold of $\mathbb{RP}^\ell$ with the Morse function $\bar{f}(x) = \sum a_i x_i^2$ are given by two cubes:*

$$M_{q,p} = S^0 \times I^{q-p-1}$$

We get that the first differentials are multiplication by

$$M_{p,p-1} = I^{q-p-1} \sqcup I^{q-p-1} = S^0 \sim \emptyset \in \Omega_0^O.$$

So the first differential is zero, hence every element $X \in \Omega_{n-p}^O$ survives. We can also see this by [Corollary 7.9](#), as we can always extend in the following way:

$$\begin{array}{ccccc} \bullet & \xrightarrow{X} & \bullet & \xrightarrow{S^0} & \bullet \\ & \searrow & & \nearrow & \\ & & X \times I & & \end{array}$$

To calculate the second differential we need to calculate the Toda brackets of the two flow categories:

$$\begin{array}{ccc} \bullet & \xrightarrow{X} \bullet & \xrightarrow{S^0} \bullet \\ & \searrow & \nearrow \\ & X \times I & \end{array} \qquad \begin{array}{ccc} \bullet & \xrightarrow{S^0} \bullet & \xrightarrow{S^0} \bullet \\ & \searrow & \nearrow \\ & S^0 \times I & \end{array}$$

Which is given by the colimit of the diagram:

$$\begin{array}{ccc} & X \times S^0 \times S^0 & \\ \swarrow & & \searrow \\ X \times I \times S^0 & & X \times S^0 \times I \end{array}$$

we get the topological space $X \times \partial I^2$, which after straightening the corners (??) is the manifold $X \times S^1$.

Proposition 7.11. *Let E be the spectral sequence of $\text{Let } \mathbb{RP}^\ell$ with the morse function $\bar{f}(x) = \sum a_i x_i^2$ and let MO be the trivial tangential structure and let $E_r^{n,p}$ be the spectral sequence arising from ?? . Then $E_r^{n,p}$ collapse at the first page and we have that*

$$MO_i(\Sigma_+^\infty \mathbb{RP}^\ell) = \bigoplus_{\ell-i \leq j \leq i} MO_j.$$

Proof. We need to show that for any $r \in \mathbb{N}$ and for any $X \in E_1$, we have that X survives to E_r .

Denote by $\mathcal{C}$ the flow category of f . By [Corollary 7.9](#), we need to extend X to a right flow module W of degree n over $\mathcal{C}_{[p,p-r]}$. We define the manifolds of W to be:

$$W_q = X \times I^{p-q}.$$

The equations

$$\partial W_q = X \times \partial I^{p-q} = \bigcup X \times I^{p-q-1} \times S^0 = \bigcup_{q < q' \leq p} X \times I^{p-q'} \times S^0 \times I_{q'-q-1} = \bigcup_{q < q' \leq p} W_{q'} \times M_{q',q}$$

justify that W is indeed a right flow module over $\mathcal{C}_{[p,p-r]}$.

Moreover, the differential is given by the Toda manifold $W \circ \mathcal{M}_{-,p-r-1}$, which is:

$$d_{r+1}(X) = \bigcup_{p \geq q \geq p-r} W_q \times M_{q,p-r-1} = \bigcup_q X \times I^{p-q} \times S^0 \times I^{q-(p-r-1)-1} = \bigcup_q X \times I^r \times S^0 = X \times \partial I^{r+1} \simeq X \times S^r \quad (47)$$

□

Now we want to calculate $MSO(\Sigma_+^\infty \mathbb{RP}^\ell)$. Under the same settings we have

Proposition 7.12. *Let E be the spectral sequence of $\mathbb{RP}^\ell$ with the morse function $\bar{f}(x) = \sum a_i x_i^2$ and coefficient ring MSO . We get that*

$$E_1^{n,p} = \Omega_{n-p}^{SO}$$

The first differential is given by multiplication with

$$d_2^{n,p} = \begin{cases} +, + & p \text{ is even} \\ +, - & p \text{ is odd} \end{cases}.$$

Moreover every element $X \in E_2$ survives to E_∞ and the reduce homology is:

$$\Omega_i^{SO}(\mathbb{RP}^\ell) = \begin{cases} \Omega_{i-\ell}^{SO}[2] \oplus \Omega_{i-\ell+1}^{SO}/2 \oplus \Omega_{i-\ell+2}^{SO}[2] \cdots \oplus \Omega_{i-1}^{SO}/2 & \ell = 2k \\ \Omega_{i-\ell}^{SO} \oplus \Omega_{i-\ell+1}^{SO}[2] \oplus \Omega_{i-\ell+2}^{SO}/2 \oplus \Omega_{i-\ell+3}^{SO}[2] \cdots \oplus \Omega_{i-1}^{SO}/2 & \ell = 2k+1 \end{cases}$$

where $\Omega_j^{SO}[2]$ are the elements $X \in \Omega_j^{SO}$ such that $2X$ is null bordant.

Proof. The existence of the spectral sequence following from [Corollary 7.9](#). The framing, hence orientation, of the zero dimensional manifolds $M_{p,p-1}$ is induces from the exact sequence:

$$T(W_p^u \cap W_{p-1}^s) \rightarrow TW_p^u \rightarrow \nu(W_{p-1}^s)$$

We denote the two components of the zero dimensional flow manifold

$$M_{p,p-1}^+, M_{p,p-1}^- \in M_{p,p-1} = [0 : \cdots : 0 : 1 : \pm 1 : 0 : \cdots : 0]$$

and we compare the framing of $M_{p,p-1}^+$ to the framing of $M_{p,p-1}^-$ by composing:

$$\varphi : T(W_p^u \cap W_{p-1}^s)_- \oplus \nu W_{p-1}^s \simeq TW_p^u \simeq T(W_p^u \cap W_{p-1}^s)_+ \oplus \nu W_{p-1}^s$$

Let $v_0, \dots, v_{p-2}$ be a basis for νW_{p-1}^s (which is of size $p-1$), the maps

$$\nu(W_{p-1}^s) \rightarrow TW_p^u$$

are induces from the parallel transport

$$\nu_{p-1}(W_{p-1}^s) \rightarrow T_p W_p^u$$

(through the points $M_{p,p-1}^\pm$) hence $\varphi|_{\nu W_{p-1}^s}$ is the holonomy along the curve

$$\gamma(t) = [0 : \cdots : 0 : \cos(t), \sin(t) : 0 : \cdots : 0], \quad t \in [0, \pi]$$

By lifting γ to the sphere we get that:

$$\varphi(v_0, \dots, v_{p-1}) = -v_0, \dots, -v_{p-2}.$$

Combine that, with the definition:

$$M_{p,p-1} := W_p^u \cap W_{p-1}^s / \nabla f$$

where the flow ∇f acts on $M_{p,p-1}^\pm$ in opposite directions, to conclude that for an element $X \in E_1^{n,p}$ we have that $d_1(X)$ is two copies of X where the framing on one copy is given by inverting p vectors comparing to the other copy. Hence

$$d_1^{n,p}(X) = \begin{cases} \bar{X} \sqcup X & p \text{ is odd} \\ X \sqcup X & p \text{ is even} \end{cases}$$

Let $X \in E_1^{n,p}$. For p odd we have that X extends to a right flow module W over $\mathcal{C}_{[p,p-1]}$ by setting:

$$W_{p-1} = X \times I.$$

As we defined the "forgetting" map $\mathbf{Flow}^{SO} \rightarrow \mathbf{Flow}^O$ induce that diagram:

$$\begin{array}{ccc} (\mathcal{E}_r^{n,p})^{SO} & \xrightarrow{d_{r+1}^{SO}} & (\mathcal{E}_r^{n-1,p-r-1})^{SO} \\ F \downarrow & & \downarrow F \\ (\mathcal{E}_r^{n,p})^O & \xrightarrow{d_{r+1}^O} & (\mathcal{E}_r^{n-1,p-r-1})^O \end{array}$$

where F is the forgetting map $\Omega^{SO} \rightarrow \Omega^O$ and $\mathcal{E}$ are the elements that survive to the $(r+1)$ -th page in the (un)oriented spectral sequence. The flow module above shows that $X \in E_1^{n,p}$ survives to E_2 , inductively, we assume that X survives to E_{r+1} (i.e., extends to a right module over $\mathcal{C}_{[p,p-r]}$). Using that diagram and Eq. (47), we get that

$$F \circ d_{r+1}^{SO}(X) \simeq d_{r+1}^O \circ F(X) \simeq F(X) \times S^r.$$

Both orientations of the circle are null-bordant, hence the underlying manifold of $d_{r+1}^{SO}(X)$ is null-bordant and X survives to E_{r+2} (i.e., the module extends over $\mathcal{C}_{[p,p-r-1]}$).

For p even, we get that $d_1(X) = X \sqcup X$, hence X survives to $E_2 \iff X$ is 2-torsion. Under this assumption, we can use the fact that d^{SO} lifts d^O to get that X survives to E_{r+1} .

We conclude that the spectral sequence collapse at the second page and

$$\Omega_i^{SO}(\mathbb{RP}^\ell) = \begin{cases} \Omega_{i-\ell}^{SO}[2] \oplus \Omega_{i-\ell+1}^{SO}/2 \oplus \Omega_{i-\ell+2}^{SO}[2] \cdots \oplus \Omega_{i-1}^{SO}/2 & \ell = 2k \\ \Omega_{i-\ell}^{SO} \oplus \Omega_{i-\ell+1}^{SO}[2] \oplus \Omega_{i-\ell+2}^{SO}/2 \oplus \Omega_{i-\ell+3}^{SO}[2] \cdots \oplus \Omega_{i-1}^{SO}/2 & \ell = 2k+1 \end{cases}$$

□

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